

Machine learning-driven insights into the microstructure and properties of high-entropy alloys

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ABSTRACT

High entropy alloys (HEAs) have recently become a popular category of alloys, composed of five or more elements. These alloys are of particular interest in the field of materials due to their unique structure and excellent properties. However, the multi-component nature of these alloys poses challenges to traditional calculation methods, necessitating the development of alternative approaches for their analysis. Machine learning, a branch of artificial intelligence, has emerged as a promising solution to address the complexity inherent in the composition and structure of HEAs. The present review focuses on the fundamental definition and process of machine learning and its application in the research field of HEAs. The primary focus of this research field is the prediction of phase structure, hardness, strength, thermodynamic properties, and catalytic properties. In addition, future perspectives on the challenges in this research area are also presented.

1. Introduction

High entropy alloys (HEAs), also referred to as multi-major element alloys [1], represent a novel class of alloys characterized by distinctive compositions, structures and properties. The structure of the material under consideration is typically comprised of a minimum of five elements [2], each existing in equimolar or specific proportions to form a chemically disordered, yet topologically ordered random solid solution structure [3]. This structure is commonly known as face-centered cubic (FCC), body-centered cubic (BCC), or hexagonal close-packed (HCP) structures. This unique structure gives rise to a variety of special effects [4], namely the high entropy effect, the lattice distortion effect, the slow diffusion effect, and the cocktail effect. In addition, the phase formation is affected by a variety of factors, including the enthalpy of mixing, the atomic size difference, and the valence electron concentration (VEC) [5]. These features contribute to the excellent performance and potential application value of HEAs in mechanical properties [6], catalysis, and energy storage.

At present, the study of the various properties of HEAs is possible through the use of materials computational methods. However, given the combinatorial explosiveness inherent in the compositional space of HEAs, traditional research methods appear overwhelming. Machine learning offers distinct advantages in the analysis of large data sets [7–12], including the high efficiency and low cost of processing datasets, as well as the ability to efficiently explore the vast combinatorial property space. Machine learning, as a key field and an important branch of artificial intelligence [13–20] can deeply integrates multi-disciplinary knowledge such as statistics, mathematics, and computer science, constituting a typical multidisciplinary cross-discipline. Its core goal is to empower computers to process and analyze huge amounts of data with the help of algorithms [21–27], to automatically mine the latent patterns, laws, and knowledge in the data, and then to make accurate predictions, reasonable classification, or scientific decisions on unknown data.

In this review, we present advances in the application of machine learning to the study of microstructure and properties of HEAs. First, we

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introduce the process of machine learning, such as data collection and feature engineering, as well as related modelling algorithms. Then, we discuss specific applications of machine learning in the study of HEAs. The discussion revolves around five areas: phase composition prediction, hardness prediction, strength prediction, thermodynamic property prediction, and catalytic property prediction. Finally, we suggest future directions for machine learning in this field, including how to address challenges such as data scarcity.

2. Process and method of machine learning

In recent years, the theory of machine learning algorithm has developed rapidly and has played an increasingly important role in many fields of scientific research. In the field of materials, machine learning uses the computer as a tool to learn the material data obtained by algorithms to obtain the potential relationship between material properties, and then uses the trained machine learning model to predict the performance of unknown materials, and can also be used to predict new materials, optimize material properties and identify the stable configuration of complex materials.

According to different learning styles, machine learning can be divided into supervised learning [32], unsupervised learning [33], semi-supervised learning [34], deep learning and reinforcement learning. As shown in Fig. 1, the process of machine learning in material research can be divided into the following four steps (Fig. 2): (1) Data collection and preprocessing: data sources are mainly from open source databases, experimental data and simulation data, etc., and the original data collected is very messy and inconsistent in format, which cannot be used directly, and needs to be preprocessed to adapt to the learning model [35,36]; (2) Feature selection: Feature: as the input parameter of the learning model, plays a very key role and has a direct impact on the

prediction results [37,38], so feature selection plays a very important role in machine learning; (3) Model selection: A good machine learning model is crucial in the research process [39,40]. According to different material research needs and data set size, choosing the most appropriate learning model can greatly reduce the computing time and cost; (4) Model construction and prediction [41]: Generally, after model construction and prediction, the model needs to be evaluated according to the obtained prediction results, in order to continuously improve the model according to the output results, so as to achieve the best prediction performance.

2.1. Data collection and preprocessing

Since machine learning analyzes the potential rules between the properties of substances based on data, and thus predicts the target properties, the quality of the data set determines the upper limit of the performance of various models and plays a decisive role [42]. A qualified data set must have the three conditions of accuracy, wide representation and sufficient scale. For HEAs, the data generally come from material databases, experimental data and first principles calculation data. For computational data, it is often used for small-scale machine learning modeling, as small-scale machine learning is often used to build alternative models of atomic interactions [42]. Moreover, the data calculated by the first principle, such as the gap width and atomic energy, are calculated according to the same parameters and methods, strictly ensuring unity, so the influence of other factors will be eliminated and the consistency of the target attributes will be guaranteed. For experimental data, databases and other data, it is generally applicable to more macro machine learning and is used as the input and output label of machine learning algorithms. For experimental data, because of its high accuracy, the input label of the model usually brings more accurate

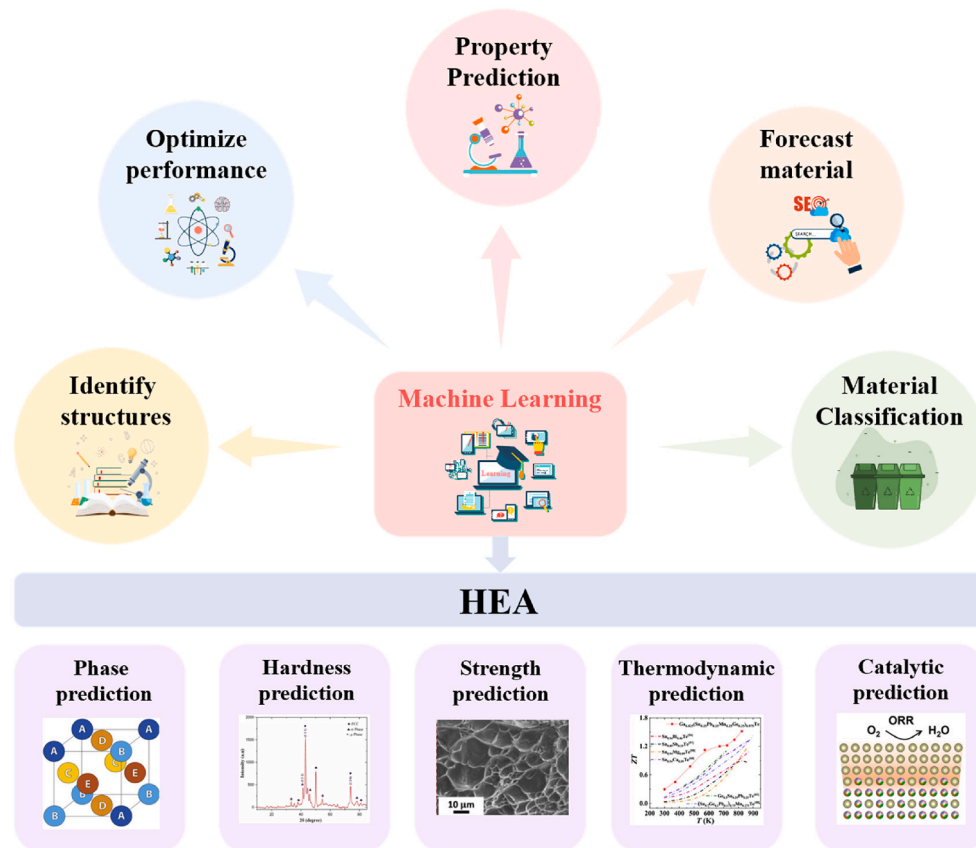


Fig. 1. The use of machine learning in the field of materials and the prediction of the microstructure and properties of HEAs driven by machine learning. Ref. [1,28–31].

prediction results. The data in the database generally comes from simulation or experimental data, and the calculation of simulation data is usually less time-consuming and the acquisition cost is low, which can be obtained in a large amount in a short time [43].

Moreover, the data in the database is easy to verify and more suitable for the study of dynamically changing HEAs. For example, in 2018, Senkov and Miracle created a database of HEAs, which included the mechanical properties of 370 refractory HEAs [44]. After obtaining the data, the more important step is to preprocess the data and select the most valuable and accurate data to build the model. Because for the same target attribute, different literatures use different characterization and preparation methods, which will cause differences in data. The purpose of data preprocessing is to improve the uniformity, incompleteness and accuracy of data, which usually includes data cleaning and standardized processing [45]. Data cleansing is usually done to remove incomplete, incorrect, or abnormal values. Normalization, on the other hand, scales the descriptor data to the interval, eliminates the measurement differences of data in different Spaces, makes the features of different scales have the same importance, and makes the data more adaptable to the model.

2.2. Feature selection

Descriptors are methods used to describe the characteristics of data samples [46], which process raw data into numerical vectors that can be processed by machine learning algorithms, such as hardness, electrical conductivity, and so on. In the process of describing material information, too many descriptors may cause overfitting. Therefore, the process of removing redundant descriptors to make the system model clear and retain more data is called feature selection [47]. Feature selection is usually divided into three methods: filtering method, packaging method, embedding method.

The filtering method mainly relies on the general characteristics of the data. The correlation between the data and the target attributes is calculated as the criterion for evaluating the importance, so as to select the best feature subset, such as the variance threshold and Pearson correlation factor [45]. It does not involve any algorithms, so it is suitable for the vast majority of learning models, easy to understand, and

very fast [48]. It is suitable for high-dimensional data. At present, these evaluation indicators have been used in a variety of material systems, for example, pearson correlation coefficient (PCC) has been applied to the identification of perovskite materials [49]. Correlation based feature selection (CFS) is applied to stress hotspot classification of FCC and HCP materials [50].

The packaging method uses a learning algorithm to evaluate the weight of a subset of features while ignoring independent measures between features and target attributes. It regards feature selection as the process of using algorithms to search and optimize features, forming many different subsets, and selecting a set of subsets with the best performance through the algorithm results, which has the risk of model dependence and overfitting, common algorithms such as support vector machines (SVM) and decision trees (DT) and so on. This method has been applied to the feature screening of crystal materials [51]. Study on structure-composition relationship of cubic perovskite oxides [52].

Although the embedding method also uses algorithms for evaluation, it imparts the feature selection process into the model training process and realizes the coordination between model construction and feature selection. Through model training, the importance of each feature is obtained and ranked, so as to select features. At present, the embedding method has been used in different fields of materials to solve a variety of problems, such as gradient lifting regression (GBR) to predict the activation energy of gap diffusion [53]. Random forest (RF) was applied to modeling the critical temperature of superconductors [54]. L1 regularization was applied to carbon fiber performance analysis [55].

2.3. Model selection

The model selection of machine learning plays a key role in the results of artificial intelligence prediction of material properties, and a suitable learning model can ensure the reliability of the final output results, and also greatly reduce the cost and time spent on training the model. A machine learning model learns training data to mine potential connections between target attributes and selected features without the need for relevant experience and special functions [56]. to help it get more accurate predictions. In the development of machine learning, a large number of model building methods have been accumulated, and

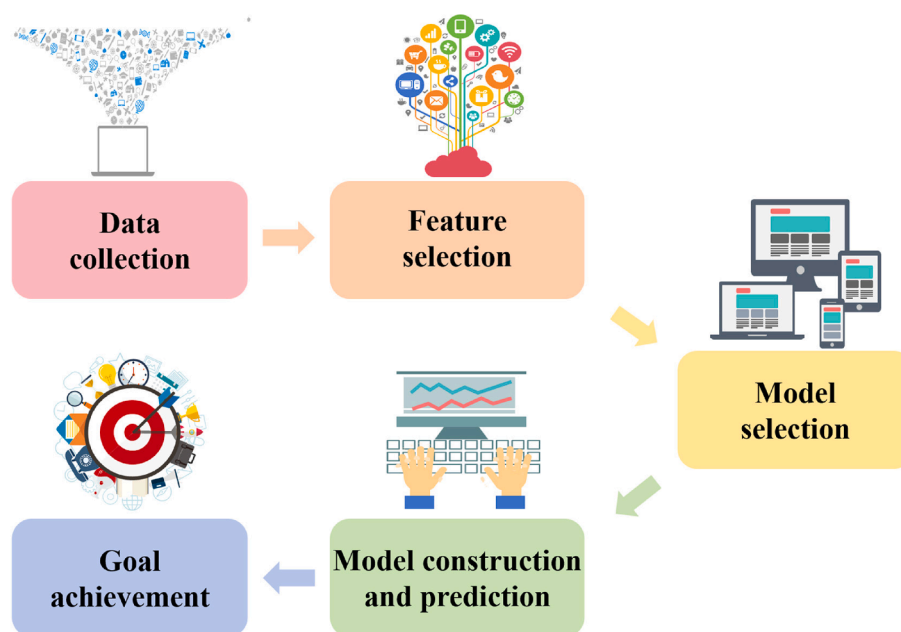


Fig. 2. Flow chart of machine learning.

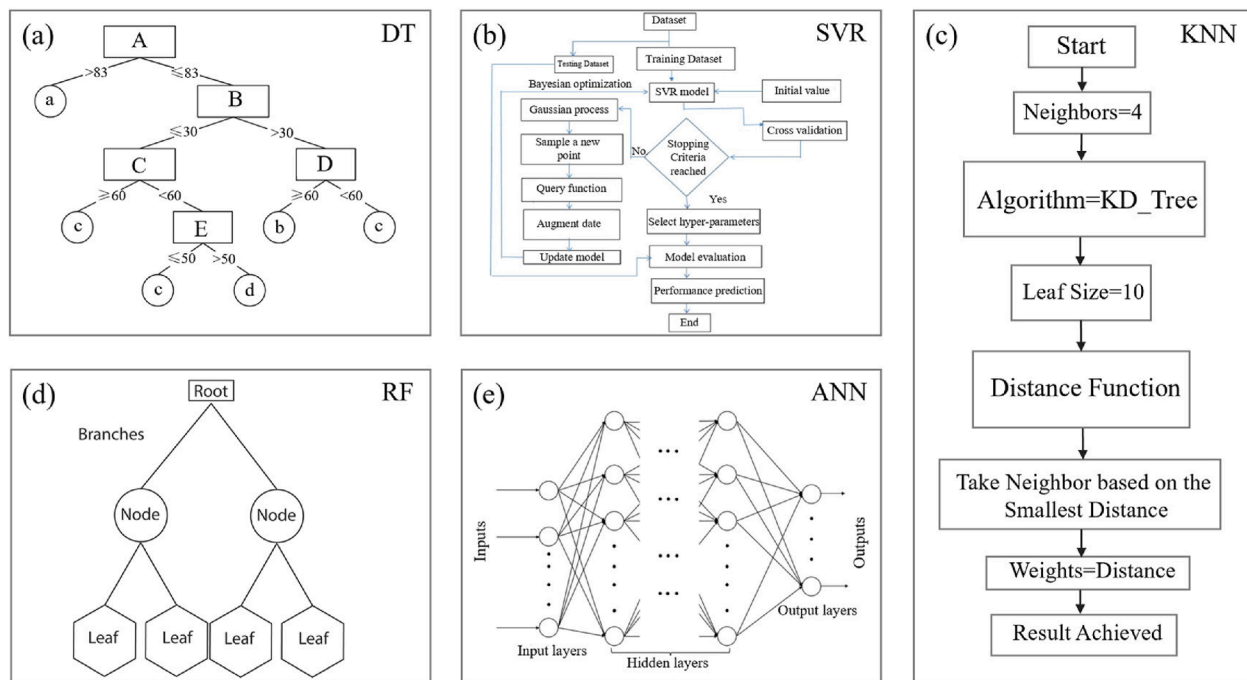


Fig. 3. (a) DT based on student sports results. Reproduced with permission from Ref. [59]. Copyright 2018, International Journal of Emerging Technologies in Learning. (b) SVR flow chart after Bayesian optimization (BO). Reproduced with permission from Ref. [73]. Copyright 2020, Journal of Energy Storage. (c) KNN frame diagram improved by adjusting parameters. Reproduced with permission from Ref. [79]. Copyright 2024, Computers, Materials & Continua. (d) A common graphical description of a RF model with roots, nodes, and leaf sections. Reproduced with permission from Ref. [80]. Copyright 2018, Chemistry of Materials. (e) Regular ANN frame diagram. Reproduced with permission from Ref. [81]. Copyright 2021, Materials Today: Proceedings.

different models cannot be selected by direct comparison. A common approach to model selection is to first select several suitable models and perform model selection by comparing their cross-validation results on specific datasets or their predictions on independent test sets [57]. Commonly used machine learning models include: DT, SVM, K-nearest neighbor (KNN), RF, artificial neural network (ANN) and so on.

DT is a prevalent algorithm that can be utilized for both regression and classification problems. Notably, its primary application is in the domain of classification problems. It employs a given data set to construct a decision tree. Each internal node in the tree corresponds to an attribute of the dataset (Fig. 3a), and each branch point represents a decision node [58]. The function of each branch point is to divide each attribute into two or more branches, a process determined by the decision rule [59]. The integration of multiple classifiers is achieved through their combination based on predefined DT models or classification requirements, thereby capturing different aspects of the features and enhancing recognition [60]. The selection of partitioning attributes can be enhanced by using information gain, information gain rate, and Gini index [61]. For example, ID3 decision tree algorithm uses information gain function as decision node [62], C4.5 algorithm uses information gain rate to remove the influence of multiple branching attributes and make up for the defect of information gain [63], classification and regression trees (CART) algorithm uses Gini index as division standard and selects attributes with small Gini index as nodes.

SVM is a supervised machine learning model that demonstrates high classification accuracy, is resistant to overfitting, and exhibits good generalization ability. Its applications include regression and classification problems, and has been successfully applied in various fields, such as biomedical [64–66] engineering science [67–70] materials science [71–73] and environmental science [74–76]. SVM is an algorithm proposed and developed on the basis of statistical learning theory (SLT) by mathematician N.V. Vladimirov after decades of research. SVM was originally used for classification problems called support vector classification (SVC), and later extended to regression problems called support vector regression (SVR) (Fig. 3b) [77]. SVM is commonly used to solve

small sample sizes, nonlinear, and high-dimensional pattern recognition problem [56]. Its core goal is to find a hyperplane that can separate the two types of sample points as much as possible, and the distance between the points and the hyperplane should be maximum, that is, the margin should be maximum [64]. If you encounter high-dimensional data and nonlinear classification problems, you need to rely on kernel functions to achieve excellent classification performances [62]. First of all, by defining a suitable kernel function, the nonlinear transformation is implemented to convert the original space to a high-dimensional space [78], and the optimal hypersurface is found as the interface in this space.

KNN algorithm is a widely used statistical recognition and classification method (Fig. 3c). It does not require training of a collected data set, but rather stores all the training data in memory using a simple algorithm [82]. This characteristic contributes to its strong flexibility, leading to its extensive use in machine learning and data mining [79,83,84]. The underlying principle of the KNN algorithm is relatively straightforward: first, the data set of known samples is input and stored, and then, an unknown sample is input. The distance between the unknown sample and all data points in the known data set is then calculated [85]. The k nearest neighbors are selected based on the calculated distance, and the category proportions of the k nearest neighbors are determined [86]. This process enables the classification of the unknown sample. The distance measurement methods employed include the Euclidean distance, Manhattan distance [87] and cosine distance [88], among others. To circumvent the influence of disparate dimensions on the proportion of features, normalization processing is adopted. The most common methods include z-score normalization and min-max normalization. In addition, the judicious selection of k value and distance measurement method is paramount in enhancing the efficacy of classification in practical applications [89].

RF is a widely utilized classification model that exhibits high prediction accuracy [90], robust generalization capability, and effective handling of outliers and noise in data [91]. The concept of random forest was originally proposed by Mr. Jeon Kum Ho in 1995 [56], and subsequently in 2001 Breiman proposed a general form of this idea, called

Random Forest [92]. A random forest comprises a multitude of decision trees (Fig. 3d), where the distribution of these trees within the forest is uniform. However, the attributes and training sets of each tree are randomly assigned [60], thereby ensuring the effective reduction of overfitting risk. The model utilizes straightforward decision rules to categorize the randomly assigned feature subsets to accomplish the objective of prediction [51]. When a target value X is entered, each tree in the forest will vote for the preferred class as the result of the prediction. It is not intuitive to choose the tree with the best performance in the forest for prediction because random forest prediction is based on the prediction results of all trees, not just one tree. RF has been extensively applied in diverse domains of materials research, including nanomaterials [93,94], crystal materials [80], material adsorption [95–97] material diffusion [98,99] and so on.

ANN is a model that processes and predicts data by simulating the operation of neurons in the human brain [100]. ANN possess a robust nonlinear processing capacity [101], rendering them well-suited for a range of regression and classification tasks. Their extensive application in the domain of materials science is well-documented [81,102,103]. The most common forms of ANN include convolutional neural networks, recurrent neural networks, and ordinary neural networks. The fundamental structure of an ANN comprises an input layer, a hidden layer, and an output layer (Fig. 3e). Each layer incorporates a function node, designated as a neuron. The fundamental principle of the ANN algorithm is to take the original data as input, with each neuron in the input layer receiving an input data [104]. The hidden layer, which is situated between the input and output layers, can comprise multiple layers. The neurons in the hidden layer process the data from the upper layer through the activation function and pass it to the next layer [105]. The primary function of the hidden layer is to extract complex features and patterns in the data, and the output layer is responsible for producing the final prediction results or classification labels [106]. The backpropagation artificial neural network (BP-ANN) is the most representative monitoring algorithm [25] in ANN, and it is superior to the traditional ANN in terms of prediction accuracy. The backpropagation algorithm is characterized by its ability to propagate errors from the output layer to the preceding layers, thereby facilitating a continuous

adjustment of weights in accordance with the output error.

Graph neural network (GNN) are algorithms that have been specifically engineered to process data that exhibits a network-like structure (Fig. 4a) [107,108]. A network can be defined as any structure composed of nodes and edges [109]. The idea is that each node absorbs information from its neighbors, just as you learn something new from a friend of a friend, and each node collects the characteristics of its neighbors, combines them with its own information [110], and upgrades its own identity labels, so that each node, or the network as a whole, has labels that are representative of its role in the network, or of the network's characteristics [111]. In material science, it can predict the material properties through the connection of atoms, and excavate the hidden relationships and laws between nodes through layers of information transfer [112], which is especially suitable for solving the problems related to relational networks. Despite the challenges posed by data scarcity and model interpretability in the application of GNN to HEAs, GNN are expected to accelerate the large-scale application of HEAs in the fields of energy [113], aerospace and other fields through the embedding of physical a priori, multimodal fusion and the construction of automated platforms, and to promote the materials science to enter the era of intelligent design [114].

Generative adversarial network (GAN) is a deep learning model trained through the adversarial game mechanism [117], the core of which consists of two neural networks (Fig. 4b): the generator is responsible for converting random noise into samples close to the true distribution of the data [118], so that the generated fake data as far as possible to be as fake as the real [119]; the discriminator is to classify the input data to determine whether the data is from the real data set, and the goal is to accurately distinguish between the real dataset and the fake dataset. During training, the two are optimized against each other by gradient backpropagation: the generator tries to make the fake data as false as possible, while the discriminator tries to accurately identify the fake data [120]. This adversarial process encourages the generator to gradually learn the distribution characteristics of the real data [121], and eventually generate highly realistic samples, while the discriminator also improves its discriminative ability in the confrontation [115]. GAN can break through the bottleneck of the traditional trial-and-error

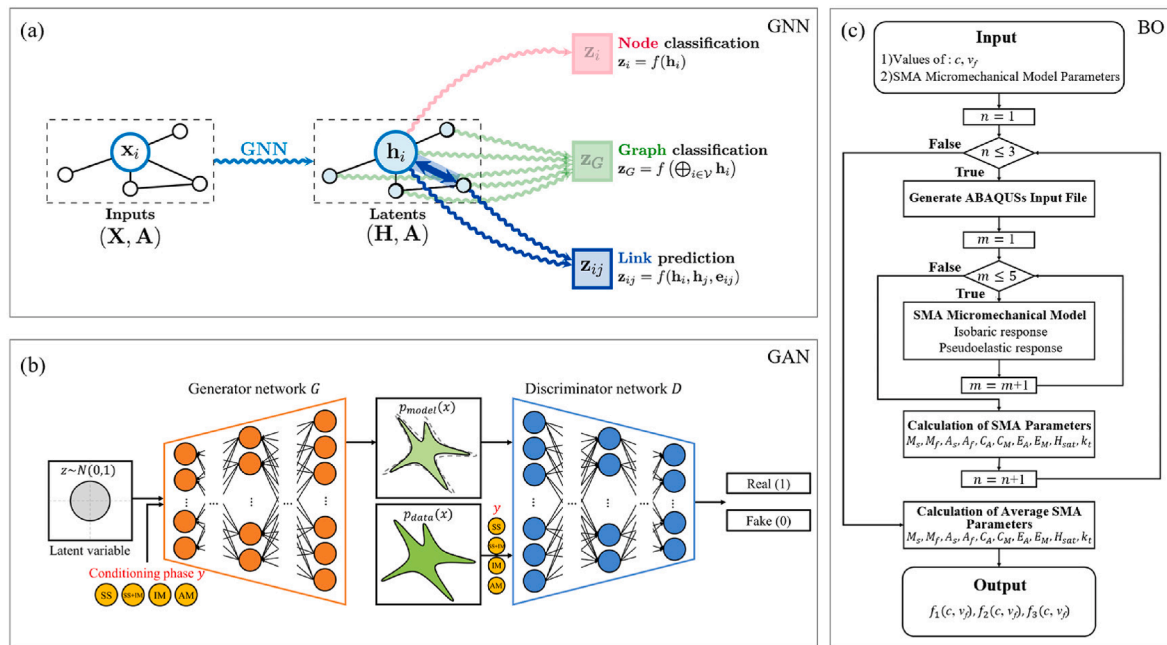


Fig. 4. (a) A common GNN architecture. Reproduced with permission from Ref. [108]. Copyright 2023, Current Opinion in Structural Biology. (b) Conditional GAN architecture with generators and discriminators. Reproduced with permission from Ref. [115]. Copyright 2021, Materials & Design. (c) A multi-objective BO design framework. Reproduced with permission from Ref. [116]. Copyright 2018, Materials & Design.

method in the design of components [122], performance prediction [123], process optimization, etc. through this competitive learning mechanism, especially for the HEAs system with huge composition combinations and high experimental costs. Despite the challenges of data quality [124], interpretability and closed-loop experimental validation, it is expected to form an intelligent design platform and accelerate the industrial application of HEAs, if it is combined with reinforcement learning, GNN and other technologies in the future.

BO is a global optimization algorithm based on probabilistic modeling, which is specifically used to efficiently find the optimal solution of complex problems [125]. When the objective function is a black box that is difficult to analyze directly [126], e.g., the cost of each experiment is high, it will first construct an agent model through Bayes' theorem with existing data (Fig. 4c), speculate where the optimal solution may be hidden, and at the same time combine with the collection function [127], e.g., expectation improvement, probability enhancement, and then decide where to go to sample next through the "Explore-Exploit" strategy [128]: Both exploring the unknown region to avoid omissions and utilizing the current optimal region to improve efficiency, iterating until the optimal point is found [129]. This method is especially suitable for solving optimization problems with limited number of evaluations and high cost, such as tuning parameter [130], experimental design, etc. It can effectively deal with high-dimensional, non-convex, and noisy optimization problems within a limited number of evaluations, minimize the optimization cost and improve the search efficiency through probabilistic modeling and adaptive sampling strategy [131]. In HEAs research and development, BO can significantly reduce the cost of trial and error [132], and can construct performance-component probability models through a small amount of experimental data to accurately recommend the most promising component combinations [115]; it can realize intelligent tuning of process parameters; and it can realize multi-objective synergistic optimization by combining with Pareto frontier analysis [116].

2.4. Model construction and evaluation

Machine learning algorithms are classified into distinct categories, with each category exhibiting optimal performance when applied to specific data sets. Typically, a selection of at least three suitable algorithms is made for the purpose of comparison. Initially, the data set is segmented into a training set and a prediction set [133], with the proportion of these sets being determined by the specific context. The proportions of these sets are typically 9:1 or 8:2. The performance of the model is evaluated by referring to the cross-validation results of different algorithms [134] or the prediction results of a specific dataset [135], usually taking root mean square error (RMSE) [136], mean square error (MSE) [137] and correlation coefficient (R^2) [138] as evaluation functions. By comparing the magnitude of error and accuracy, the algorithm with the smallest and most accurate error is selected [139].

Subsequent to the construction of the model, it is imperative to optimize the model. That is to say, one must consider the impact of feature sets and hyperparameters on the model's predictive ability [133]. In the context of features, it is imperative to deliberate on the influence of feature quantity and feature subset on the model. This entails the assessment of feature importance and the determination of the optimal feature number [140]. Such an approach has the potential to enhance the model's prediction capability while concurrently reducing computational expenses. For hyperparameters, common optimization methods include grid search [141], BO [142] and genetic algorithm (GA) [143]. In the process of hyperparameter optimization, cross-validation is frequently employed to assess the predictive performance of the model corresponding to a specific hyperparameter combination [144]. This approach enables more precise estimation of the model's performance on unseen data, facilitating the selection of the optimal hyperparameter combination.

In summary, the objective of model evaluation is to enhance the

capacity to predict target attributes, thereby improving the accuracy when encountering unknown samples [145]. It is imperative to acknowledge that the evaluation value of a model does not necessarily signify its overall performance. Instead, it offers a limited reflection of its capabilities. Consequently, the selection of diverse evaluation indicators for a model is paramount, contingent upon the particular requirements and scenarios, with the objective of minimizing prediction errors [146].

3. Application of machine learning in the study of HEAs

As a novel material concept that has gained popularity in recent years, HEAs exhibit numerous superior properties in comparison to traditional alloys. These properties include corrosion resistance [147], high mechanical properties [148] and wear resistance [149]. However, due to the intricacy of their composition and structure, the search space is considerably extensive [150]. Conventional calculation methods are inadequate due to their low practical efficiency and high cost [151]. However, recent advancements in machine learning have emerged as a promising solution, offering high efficiency, accuracy, and cost-effectiveness in predicting the properties of HEAs [152]. This method has seen widespread adoption in the field of HEAs. This encompasses the prediction of phase structure, strength, and hardness, among other properties (see Table 1).

3.1. Phase structure prediction

The composition and spatial structure of HEAs are complicated [1, 153–157], resulting in the existence of numerous phase types. The classification of HEAs phases varies across studies, including solid solution (SS), intermetallic compounds (IM), and mixed phase [158]. For different phases, the microstructure and mechanical properties are very different [153], so in order to be able to obtain the desired properties of the alloy, the prediction of the phase structure is the most critical and the most important [159]. Machine learning algorithms have emerged as a highly effective research method for predicting the phase structure of HEAs.

Islam et al. [160] used Pandas Library to import 118 instances and 6 features as a dataset, but the 6 feature values contained not only common numerical features such as *VEC*, but also a classification feature that represented different stages including amorphous (AM), etc. Then, by using the Rectified linear unit (ReLU) activation function in the hidden layer, they trained a neural network model (NN) (Fig. 5a) to make classification prediction for IM, SS and AM, and through the characteristic information provided by the model matrix, we can see that the *VEC* has the largest importance ranking in phase prediction, and the entropy change is the smallest. After prediction, the accuracy rate of the model on the training set was greater than 99 %. In order to verify the generalization ability of the model, the method of four-fold cross-validation was used to classify the test set, and the average accuracy rate of the result was only 83.0 %, indicating that the model had overfitting phenomenon, the reason may be that the data set was too small. Prediction performance can be improved by using larger data sets. In this study, although high accuracy of high-entropy alloy phase classification is achieved by NN, the dataset size is too small and overly relies on static component descriptors. In the future, we can construct a dataset with thousands of samples containing multiple processes and introduce process parameters such as cooling rate to improve the accuracy.

In addition, only one model is used for prediction, and the accuracy of the results cannot be guaranteed without numerical comparison. Therefore, some other models can be used for comparison, so as to select the model with the best performance. Huang et al. [161] collected experimental data from review articles and processed them to obtain a dataset consisting of 401 data pieces and five common characteristics such as *VEC*. They classified the dataset using three different algorithms: KNN, SVM, and ANN, and tested it using a quadruple cross-validation method to avoid overfitting the predictions. For KNN model, no matter how the parameters are adjusted, the test accuracy is lower than 70

%, because the classification performance of KNN model will be affected by the uneven distribution of data, so it is not suitable for classification problems. For SVM and multi-layer feed-forward neural network (MLFFNN), the accuracy of ternary classification is not too high, because the boundary between SS and SS + IM phase is fuzzy, so it is converted to binary classification: In SS and IM stages, SS+ IM and IM stages, and SS and SS+ IM stages (Fig. 5b), it was found that the accuracy of MLFFNN reached 86.7 %, 94.3 % and 78.9 %, respectively, which were higher than the SVM model, so it was concluded that MLFFNN was the best model. In this study, although the phase prediction accuracy was improved by multi-algorithm comparison, the over-reliance on traditional descriptors resulted in the model's inability to distinguish the phase differences between different processes under the same composition. A joint composition-process descriptor can be developed to capture the parameters under different processes.

Further improving the predictive accuracy of the model is where we should continue to improve in the future. Li et al. [162] collected experimental synthesis data from review articles and processed them to obtain 322 data sets. They chose support vector machine as the model, and carried out forward feature search based on wrapper, and then placed the alternative features into the feature set one by one. It was found that when the sub-set contained five features, VEC, atomic radius(r_d), melting point (T_m), entropy change (ΔS) and high configurational entropy(S_C) (Fig. 5c), the prediction accuracy was the highest, and more features would easily lead to overfitting of the model. By using this feature set and an optimized SVM model to make classification predictions for FCC, BCC,

and NSP phases in the dataset, they achieved a training accuracy of up to 96.55 %. In addition to data classification, the SVM model can also be used to predict new HEAs. We selected a composition space containing 16 elements and successfully predicted 369 face-centered cubic and 267 body-centered cubic diatomic HEAs, which is twice as large as the existing experimental data. At the same time, several dozen kinds of insoluble HEAs were selected based on density and melting temperature (T_{melt}), and 11 of them were verified by experiments. In addition, 20 kinds of five-element alloys with the highest T_m were also verified by first principles. In this study, the SVM model significantly improves the phase prediction efficiency of HEAs, but it is limited by the small sample size in the dataset and the absence of the dynamic parameters of the preparation process, which may bias the prediction of complex systems. It is suggested that the data of rare phases and multi-component alloys can be expanded, and the preparation parameters can be integrated.

The number of selected features also has an impact on the predicted value, in addition to the size of the data set and the model selection. For example, Risal et al. [163] used the data collected from the references as the data source and selected six features for the prediction of phase classification, which is one more feature than the previous study of Huang et al. [161]. These characteristics are VEC, electronegativity difference ($\Delta\chi$), atomic scale difference (δ), ΔS , enthalpy change (ΔH), and T_{melt} (Fig. 5d). After feature selection by two different methods, entropy change is the least important feature, but after principal component analysis and correlation analysis between features, all features are retained. Firstly, nine data sets were obtained after the data

Table 1

A summary of this review on the application of machine learning in the microstructure and performance of HEAs.

Algorithms	Dataset	Target property	Size	Performance	Year	Refs
	Key features					
NN	VEC, $\Delta\chi$, δ , ΔH , ΔS	AM, IM, SS	118	Average accuracy = 83 %	2018	[160]
MLFENN	VEC, $\Delta\chi$, δ , ΔH , ΔS	SS, IM, SS + IM	401	Accuracy = 86.7 %, 94.3 %, 78.9 %	2019	[161]
SVM	S_C , ΔH , T_m , VEC, r_d	FCC, BCC, NSP	322	Accuracy $\geq$ 90%	2019	[162]
KNN, RF	VEC, $\Delta\chi$, δ , ΔH , ΔS , T_m	SS, IM, SS + IM	598	Accuracy = 92.31 %, 91.21 %	2021	[163]
SVM, Nnet	$\overline{AN}$, δ_{XMB} , δRC , δTB ; $\overline{VEC}$, $\overline{DCE}$, δ_{XP} , δMC	FCC, BCC, DP	550	Accuracy = 88.7 %, 91.3 %	2020	[164]
XGBoost	VEC, δ , T_m	HV, Friction coefficients	184	$R^2 = 0.82$	2024	[169]
RF	E_i^{3rd} , E_i^{1rd} , δK , δG ...	HV	261	$R^2 = 0.91$	2023	[170]
SVR-R	VEC, ΔS , e/a , γ_s	HV	205	RMSE = 47.754	2022	[171]
SVR	VEC, δ , $\bar{r}$, T_m , ΔH , ΔS	HV	370	RMSE = 10.3	2024	[172]
LR, KNN, ANN	T_l , RM , ΔH , ΔS , EM ...	HV	739	$R^2 = 0.944$	2024	[173]
RFR, DNN, XGBoosting, SVR	Ti, F-A-Time, Fe, Al, C-R ...	HV, Phase volume fraction	232	Accuracy $\geq$ 89%	2024	[174]
RF	VEC, B , ΔS , T_m	Yield strength	240	Accuracy $\geq$ 93%	2021	[182]
GB	ΔS , δ , $\Delta\chi$, γ_{us} , $\bar{X}_l$, ΔH	Yield strength	463	$R^2 = 0.88$	2023	[183]
AL	ΔS , δR , TIE ...	Compressive specific yield strength	99	–	2024	[184]
Auto Gluon Tabula	–	Yield strength	–	–	2023	[185]
RF	ΔS , μ , ΔH , δ_{VEC} , $D.G$, W^6 , $ADTUS$, $ADDEN$, E_C	Yield strength	224	$R^2 = 0.942$	2024	[186]
RF, KNN	Strain, Temperature, Strain rate	Flow stress	1730	$R^2 = 0.97$, 0.958	2024	[191]
XGBR	H , C_{ym} , C_a , a , C_{en} , v_{ec} , C_c , c	M_F , T_{hy}	108	$R^2 = 0.902$, 0.928	2024	[150]
GBT	$\overline{VEC}$, $\Delta\chi$, ΔS , $\bar{N}$, e/a , f_f	Ms	371	$R^2 = 0.92$	2024	[192]
CE	T_{do} , T_{in} , T_{ho} , EFA , $DEED$, δ , ΔE_f , ΔE_{hull}	Structural (meta-) stability	2482	$R^2 = 0.94$	2024	[193]
NN	GN , PN , EN , N_{ie-d}	Adsorption energy	1280	Mean MAE = 0.086	2022	[198]
EML	$j_{relative}$	HER activity	238	Mean $R^2 = 0.93$	2024	[199]
GNN	O^* , OH^*	Adsorption enthalpies	19955	$MAE \leq 0.1eV$	2024	[200]
GCN	Number of on top *OH ΔE_{DFT} , FCC hollow *O ΔE_{DFT}	Adsorption energy	–	$MAE = 0.058$, 0.066	2023	[201]
NB, GBR	Zn, Co, Mg ...	T_{90}	381	F1 = 0.95	2024	[202]

Note: Atom number: $\overline{AN}$, Mismatches in Martynov and Batsanov electronegativities: δ_{XMB} , Covalent radius: δRC , Boiling temperature: δTB , Core electron distance: $\overline{DCE}$, Mismatch in Pauling electronegativities: δ_{XP} , Mismatch in compression modulus: δMC , Neural Network classifier: Nnet, First ionization energy: E_i^{1rd} , Bulk modulus: δK , Shear modulus: δG , Number of itinerant electrons: e/a , Cold rolling: C-R, First aging time: F-A-Time, Unstable stacking fault energy: γ_{us} , Weight average property: $\bar{X}_l$, Third ionization energy: TIE , Effect of the atomic-size difference: δR , Lattice distortion energy: μ , Standard deviation of the valence electron concentration: δ_{VEC} , Shear modulus mismatch: $D.G$, Sixth power of the work function: W^6 , Average deviation of the unfilled total valence electrons for the composition: $ADTUS$, Average deviation of the density: $ADDEN$, Average cohesive energy: E_C , Enthalpy of mixing: H , Deviation of ym: C_{ym} , Deviation of a: C_a , Lattice constant: a , Deviation of en: C_{en} , Valence electron numbers: v_{ec} , Deviation of c: C_c , Lattice constant c: c , Mean number: $\bar{N}$, Frac f valence electrons: f_f , Entropy-formation ability: EFA , Disordered enthalpy-entropy descriptor: $DEED$, Formation energies: ΔE_f , Enthalpy contributions: ΔE_{hull} , Group number: GN , Period number: PN , Electronegativity: EN , the number of unpaired electrons in d orbitals: N_{ie-d} , Adsorption energies: ΔE_{DFT} .

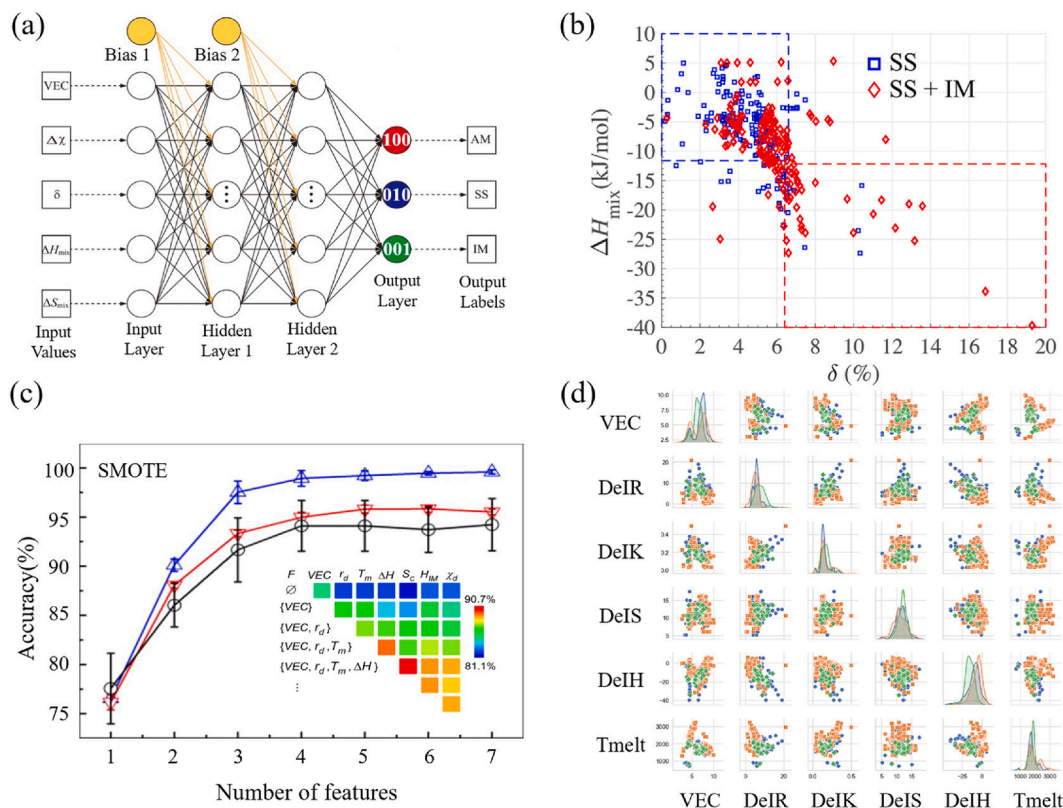


Fig. 5. (a) Structure of a NN model containing two hidden layers. Reproduced with permission from Ref. [160]. Copyright 2018, Computational Materials Science. (b) A scatter plot containing the labels of the phases in 347 HEAs. Reproduced with permission from Ref. [161]. Copyright 2019, Acta Materialia. (c) Forward feature search process; Test accuracy of the original dataset and the SMOTE dataset during the feature selection process. Reproduced with permission from Ref. [162]. Copyright 2019, Physical Review Materials. (d) Scatterplot of input features for three phases. Reproduced with permission from Ref. [163]. Copyright 2021, Computational Materials Science.

collected by the literature were preprocessed and sampled. Then, four different algorithms were used to make classification prediction of the data set. Among them, KNN and RF had the highest prediction accuracy in the oversampled PCA-6 data set, reaching 92.31 % and 91.21 % respectively. For SS + IM and SS, the classification accuracy reached 81.7% and 85.1 % respectively. In this study, phase prediction accuracy is improved through multi-algorithm comparison and data preprocessing, but over-reliance on thermodynamic parameters may lead to insufficient generalization of predictions for complex alloy systems. It is recommended to introduce atomic-level features and high-throughput experimental data, and to strengthen the small-sample learning technique to solve the problem of rare phase data scarcity.

Beyond the above research methodologies, combining learning models and feature subsets for screening can be a powerful way of building machine learning frameworks. Because there are too many available models and features, it takes time to select them separately, so finding a method that can quickly screen out the best combination of the two can effectively improve the efficiency of prediction. By constructing a framework (Fig. 6b), Zhang et al. [164] used GA to filter the combination of learning models and feature subsets from a not too large selection space, and used the optimized model to make predictions. They collected 550 data from different literatures and divided them into two categories, and collected 70 features related to phase prediction, which were combined and screened by GA, and the accuracy of classification I and classification II reached 88.7 % and 91.3 %, respectively, which was more than 10 % higher than the accuracy of traditional methods. In addition, they also used active learning (AL) (Fig. 6c) to improve the accuracy of the model, and the experiment has been verified. Although this study improves the phase prediction accuracy of HEAs by GA and

AL, it only focuses on the composition-phase structure correlation and completely ignores the key role of processing and microstructural features. In the future, we can construct an all-link prediction framework that is closer to the actual material design process, and promote the evolution of multi-physics collaborative optimization.

HEAs are composed of many different components, and the phase structure of different components is different, so many studies on HEAs start from the phase composition prediction. Because of the different phase composition, the microstructure, mechanical properties and thermodynamic properties of the alloy are different. At present, many methods for predicting phase composition and structure have been perfected, and we hope to see the emergence of more accurate, simpler and less costly methods in the future.

3.2. Hardness prediction

HEAs demonstrate exceptional structural and functional characteristics, attributable to the diversity of their constituent elements. The interaction among these elements leads to an increase in the mixing entropy of the alloys, thereby exerting an influence on mechanical properties, such as hardness [28,165–168]. This attribute endows them with considerable application prospects in specific mechanical manufacturing domains, including aircraft, automobiles, and industrial manufacturing materials. In a multitude of significant industrial materials, hardness exerts a substantial influence on performance during utilization. Consequently, the prediction and design of hardness are indispensable for the successful development of HEAs with the desired mechanical properties.

Gao et al. [169] used a machine learning model to predict the phase

evolution and hardness of cobalt-free five-element alloys, while exploring the effects of Cr on the phase transition and hardness of the alloys. First, they collected data from other research articles and obtained a dataset containing 184 data after processing. Then, considering the influence of data set size on model prediction performance, they selected three algorithms suitable for small data sets, including RF, KNN and extreme gradient boosting (XGBoost) model. It is optimized by grid search and ten-fold cross-validation to select the best model. At the same time, the importance of the selected features is ranked, the VEC, shear modulus (G) and shear modulus difference are selected as the input features of the hardness prediction model, and the mean square error and R square are selected as the evaluation parameters. For the prediction of phase evolution, KNN cross-validation accuracy is 0.761 and RF is 0.739, which is more advantageous than KNN. For hardness prediction, the accuracy of the optimized XGBoost model reaches 0.836, while the accuracy of RF is 0.831, which is slightly lower. At the same time, RF model evaluation results are also worse than XGBoost, so XGBoost model has better predictive performance. The predicted results show that when $x \leq 0.5$ in $\text{AlCr}_x\text{CuFeNi}_2$ alloy, it is a single-phase face-centered cubic structure, when $x \geq 4.25$, it is a single-phase body-centered cubic structure, and with the increase of Cr content, the hardness also increases gradually, and the deviation from the predicted value is between 5.4 and 37.3HV. In this study, the performance of cobalt-free HEAs is verified by combining machine learning models and experiments, but the dataset is too small, resulting in limited model generalization ability. It is suggested that high-throughput experimental data across the component space can be expanded to improve the robustness of prediction.

In addition to element content, the size of the selected dataset also affects the accuracy of the model. Gao et al. [170] used machine learning methods to construct an efficient and accurate model based on solution strengthening mechanisms to predict the hardness of HEAs. First, they

collected 261 hardness data from the literature and published experiments, which were preprocessed to form a dataset. Secondly, 32 features were selected to form a feature pool to identify the hardness characteristics of the alloy. However, considering that too many features will cause the model to overfit, the correlation coefficient is used to screen the features, and 72 feature subsets are obtained. Then, the data set is divided into the training set and the test set according to the ratio of 85:15, and the hyperparameters of the selected five different algorithms are optimized using the ten-fold cross-validation method. Finally, RMSE, mean absolute error (MAE), relative absolute error (RAE) and R^2 were used to evaluate the model. The results of the prediction show that all five models perform well on the training set, but the generalization ability is also very important for a model. On the test set, the prediction results of RF and SVM are more accurate (Fig. 7a), and the comprehensive performance of RF is more prominent, and the correlation coefficient R^2 is 0.91 (Fig. 7b). In order to study the influence of different features on the model, shapley additive explanationsa (SHAP) method is used to rank the importance of RF features, and the results show that the third ionization energy (E_I^{3rd}) is the most important feature. Third, the smaller the ionization energy is, the more easily the atoms become cations and form defects in the alloy, thus impeding the dislocation slip and increasing the hardness of the alloy. In this study, the hardness prediction of HEAs was realized based on the RF model, but the 261 data were insufficient for the spatial coverage of complex compositions, which may lead to the bias of extreme composition prediction. It is recommended to extend the dataset by combining high-throughput first principles calculation to improve the model's adaptability to multivariate complex systems.

In order to achieve higher prediction accuracy, it is imperative to select the most suitable machine learning models and to combine features in a rational manner. Li et al. [171] propose the utilization of an enhanced GA for the purpose of feature selection, a strategy that has

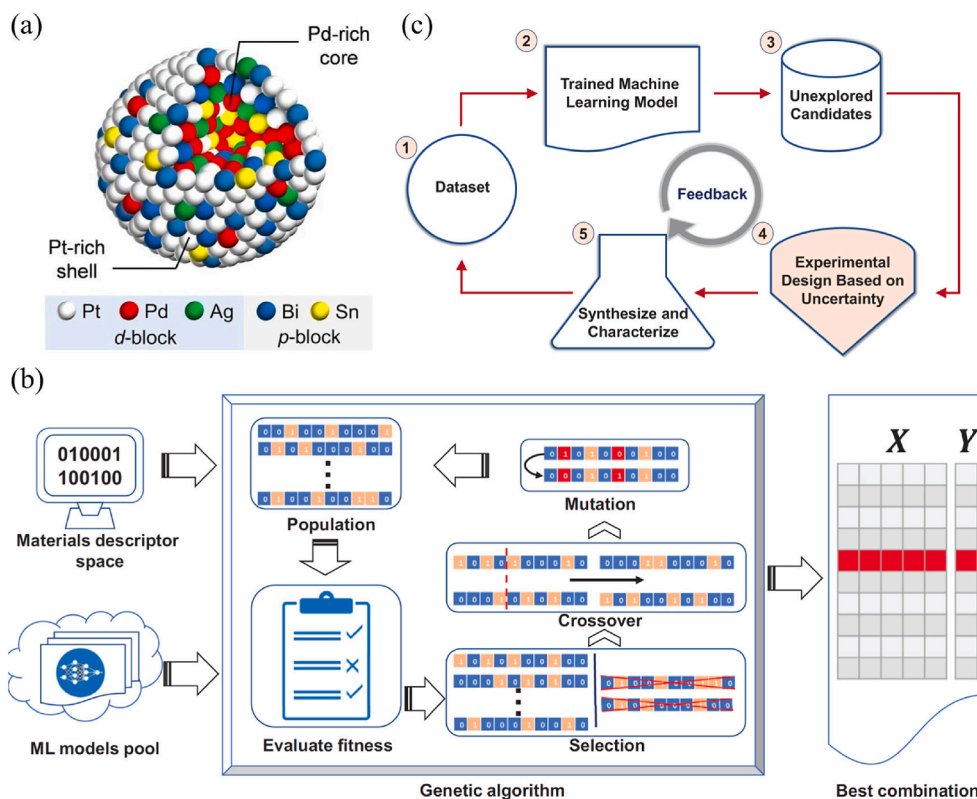


Fig. 6. (a) Microstructure of a HEAs. Reproduced with permission from Ref. [157]. Copyright 2023, Angewandte Chemie International Edition. (b) Flowchart for filtering the best combination of learning models and feature subsets and (c) an AL flowchart. Reproduced with permission from Ref. [164]. Copyright 2020, Acta Materialia.

been shown to enhance the efficacy of the model. The GA process involves the generation of initial clusters, the calculation of fitness, the selection of features, the implementation of crossover and mutation. The conventional GA (GA-0) method generates clusters randomly and iteratively optimizes a subset of features through roulette selection, crossover, and mutation. The enhanced GA (GA-1) method generates clusters based on the importance of features, adjusts the selection strategy, and selects mutated genes. GA-2 and GA-3 differ from GA-1 in their approaches to generating and selecting initial populations, respectively. The first method involved the collection of data sets from existing literature. The second method entailed the selection of several common models, leading to the selection of the SVR-R model through a comparative analysis. The optimal number of features was determined to be four (Fig. 7c), based on the prediction error. GA-1 exhibited the highest accuracy, stability, and efficiency among the compared GA algorithms. The optimal feature subsets were calculated as VEC, ΔS , atomic size mismatch (γ), and number of itinerant electrons (e_1/a), among which VEC and ΔS contributed the most to the predicted values and indirectly affected the hardness. Finally, the improved model was employed to predict the hardness of Al-Co-Cr-Cu-Fe-Ni HEAs. The predictive performance of the support vector regression with a radial basis function kernel (SVR-R) model was evaluated using ten-fold cross-validation, revealing a satisfactory outcome (Fig. 7d). Additionally, a stacking method was proposed to further minimize the prediction error of the model. In this study, the feature selection efficiency for HEAs hardness prediction is enhanced by improving the GA algorithm, but only focusing on the Al-Co-Cr-Cu-Fe-Ni system. Moreover, the model evaluation relies on cross-validation and lacks independent experimental validation. It is recommended to extend the dataset to multivariate systems, and to provide a basis for algorithm optimization by experimentally measuring the hardness and comparing the prediction deviations.

However, the reliability of the prediction results using machine learning models needs to be improved, which can be verified by experiments. Catal et al. [172] used machines learning to design a novel

refractory high entropy alloys (RHEAs) with good ductility at room temperature. First, they collected 370 data from other literature and divided them into four new datasets according to their nature. In addition, in order to be able to mine potential relationships between the data, six fundamental characteristics were added: VEC, atomic size difference, average atomic radius, T_m , enthalpy of mixing, and entropy of mixing. They then selected three different machine learning algorithms, namely SVM, ANN, and RF. According to their prediction accuracy and training error on the training set and test set, SVR model is suitable for hardness and ductility prediction, while ANN is suitable for strength prediction at high temperature and room temperature. Three optimal new HEAs $\text{Ti}_8\text{Nb}_{21}\text{Zr}_{27}\text{Ta}_{13}\text{Mo}_{19}\text{V}_{12}$, $\text{Ti}_{10}\text{Nb}_{19}\text{Zr}_{15}\text{Ta}_{43}\text{Mo}_7\text{V}_6$, and $\text{Ti}_{10}\text{Nb}_{20}\text{Zr}_{37}\text{Mo}_{21}\text{V}_{12}$ were obtained from the predicted results (Fig. 8a), and were verified by experiments. All three new alloys have good high temperature strength and room temperature ductility, especially $\text{Ti}_{10}\text{Nb}_{19}\text{Zr}_{15}\text{Ta}_{43}\text{Mo}_7\text{V}_6$ alloy can resist thermal softening at 800 °C (Fig. 8b–c). In this study, although RHEAs were successfully designed with the help of machine learning, the machine learning model did not incorporate microstructural information and the high-temperature strength prediction error was large. The construction can be followed by adding microstructural features to the model training and supplementing the actual melting point data of the alloy to optimize the high-temperature performance prediction model.

By implementing the stacked integrated learning method in the training of the multi-classification model, Chen et al. [173] enhanced the model's accuracy in predicting new alloy systems and the efficiency of alloy design. Additionally, the model demonstrated favorable generalization performance with high accuracy. Initially, a comprehensive collection of experimental data from extant literature was undertaken to establish a dataset encompassing 739 entries concerning alloy composition and hardness. Subsequently, a classification of the features was conducted to enhance the generalization ability. The SHAP values demonstrate that the third ionization energy (TI), metal radius (RM), and ΔH features are of the utmost importance. Subsequently, the integrated models EN1 and EN2 were constructed, and EN2 was subjected to

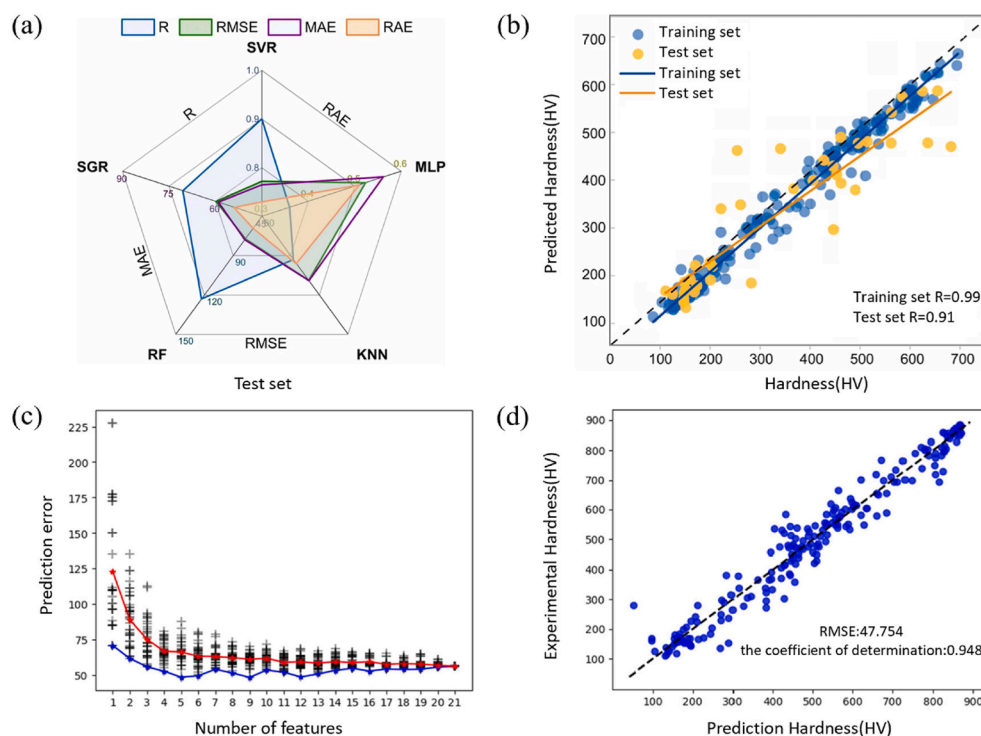


Fig. 7. (a) Comparison chart of prediction accuracy of different machine learning models and (b) prediction error of RF model on test and training sets. Reproduced with permission from Ref. [170], Copyright 2023, Materials Today Communications. (c) The prediction error is plotted against the number of features. Each point is a subset of features; the red line shows the mean error value and the blue line shows the lowest error value and (d) The relationship between predicted and measured hardness values based on the ten-fold cross-validation results of the SVR-R model. Reproduced with permission from Ref. [171]. Copyright 2022, Computational Materials Science.

testing, resulting in higher prediction accuracy ($R^2=0.944$). Subsequently, the ADASYN algorithm was employed to address the issue of sample imbalance, and the artificial neural network classifier (ANNC) model was selected to serve as the phase structure prediction model. The EN2 model was employed to predict the hardness of Al-Ti-V-Nb-Zr alloys, and two candidate alloys were selected for sample preparation. The hardness of $\text{Al}_{0.38}\text{Ti}_{0.36}\text{V}_{0.05}\text{Zr}_{0.16}\text{Nb}_{0.05}$ was measured to be 723.7 HV, and the hardness of $\text{Al}_{0.51}\text{Ti}_{0.28}\text{V}_{0.04}\text{Zr}_{0.16}\text{Nb}_{0.01}$ was measured to be 691.0 HV, with an error of less than 8 %. In this study, the prediction accuracy of the hardness of HEAs is significantly improved by stacked integrated learning, and the reliability of the model is experimentally validated. However, the experimental validation was only for two alloys with a small sample size. It is recommended to further expand the scope and diversity of the dataset in the future to improve the generalizability and robustness of the model.

A body of research has previously explored a particular property of the alloy; however, there is a correlation between various properties of HEAs. Exploration of the comprehensive properties of the alloys will lead to enhanced performance. Li et al. [174] designed a comprehensive prediction system using machine learning and used it to predict the comprehensive properties of $\text{L1}_2\text{Fe-Co-Ni}$ based HEAs to design new HEAs with superior performance. Initially, they collected data for processing to generate a dataset. Subsequently, they selected algorithms, employing random forest classifier (RFC), decision tree classifier (DTC), and SVC in the first stage; deep neural network (DNN), random forest regression (RFR), XGBoosting, and SVR in the second stage. They employed entropy weighting to combine the algorithms with the combined values of precipitation volume fraction and hardness of the alloy L1_2 for prediction. In the third stage, the algorithms from the second stage were combined with the yield strength, tensile strength, and elongation variations of the alloy to predict the combined strength and plasticity of the HEAs. The prediction results demonstrate that the

accuracy of the model is enhanced through the filtering out of irrelevant features in the first stage. In the second stage, XGBoosting provides the most accurate prediction with over 89 % accuracy. In the third stage, XGBoosting again demonstrated the highest level of accuracy, reaching over 92 %. The three optimal models were then integrated to establish a comprehensive prediction system for the screening of HEAs. This approach yielded four alloys that satisfied the specified criteria, which were subsequently subjected to experimental verification. In this study, multi-dimensional optimization of HEAs from microstructure to macroscopic performance is achieved by machine learning methods. However, the limited size of the dataset may affect the generalizability of the model. It is recommended that the size and diversity of the dataset be further expanded in future studies, and the AL strategy be introduced to dynamically optimize the data collection, so as to more comprehensively improve the performance of the model.

Eutectic high entropy alloys (EHEAs) are a class of materials that exhibit a combination of advantageous properties derived from the eutectic and high entropy phases. The high hardness of EHEAs has been demonstrated to offer enhanced wear resistance, a critical property in many applications. In their study, Lan et al. [168] utilized selective laser melting (SLM) to prepare $\text{AlCoCrFeNi}_{2.1}$ EHEAs and examined their high temperature tribological properties. Initially, the samples were prepared using SLM, employing multibeam laser strip scanning. Secondly, a series of experiments were conducted to assess the coefficient of friction under various conditions. These conditions included different temperatures, specific loads, sliding speeds, and test cycles. The volume loss and wear rate were also calculated. The hardness of the prepared alloys exhibited a gradual decrease up to an approximate temperature of 400 °C, followed by a rapid decline to approximately 105 HV at 1000 °C. The fluctuations in the coefficient of friction (COF) exhibited a direct correlation with the size of the wear debris, demonstrating an incremental rise from 20 °C to 400 °C, a subsequent decline from 600 °C to 800 °C,

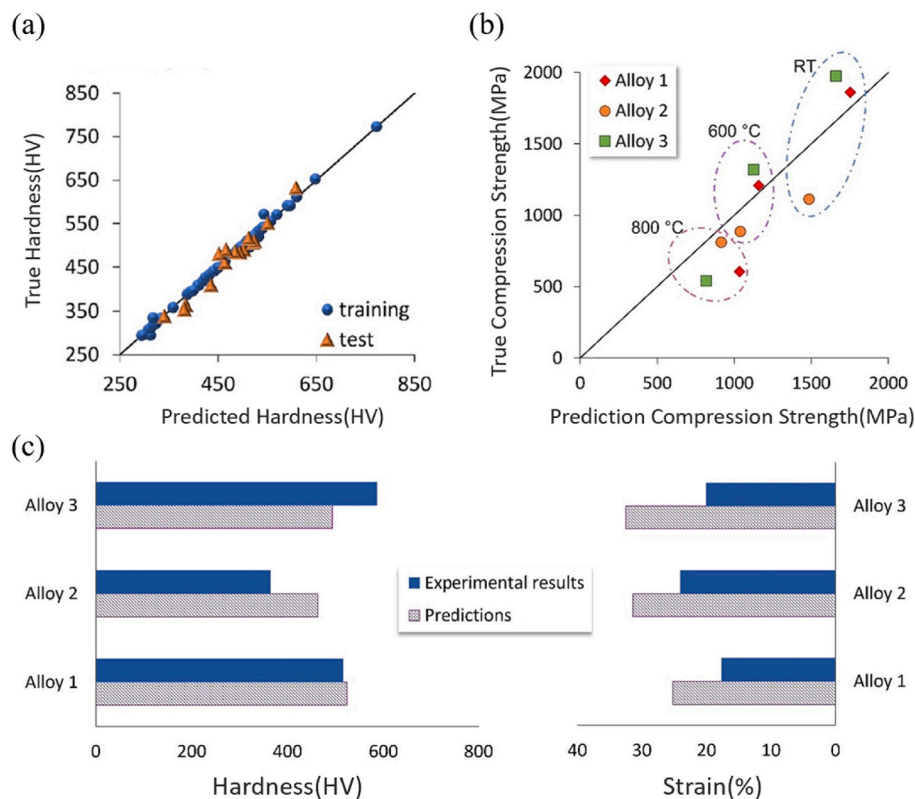


Fig. 8. (a) Comparison of predicted and true values of the hardness dataset based on the ANN model; (b) Comparison of actual and predicted compressive strengths for alloys 1,2,3; (c) Plot of experimental results versus predicted values for hardness and strain for alloys 1,2,3. Reproduced with permission from Ref. [172]. Copyright 2024, Computational Materials Science.

and a modest fluctuation with delamination occurring at 1000 °C. The wear rate exhibited an upward trend in conjunction with elevated temperatures, as well as specific loads and sliding speeds. The wear rate exhibits an increase followed by a decrease in relation to the temperature, and a transition is observed at 400 °C. The findings indicate that the wear rate of the wear chips exhibits an increase with rising temperature, followed by a decrease with decreasing temperature. A thorough examination of the experimental findings reveals that as the temperature rises, the simple oxides undergo a reaction, transforming into complex oxides. This transformation is accompanied by an enhancement in the wear resistance of the dense bimetallic oxide layer. Furthermore, during the friction process, various effects, including frictional heat and surface physiochemistry, act in a synergistic manner to influence the chemical reaction and wear process at the friction interface. This study presents a comprehensive set of experimental data and theoretical analyses, offering a comprehensive understanding of the performance of HEAs in high-temperature friction environments. The findings of this study contribute to the expansion of the application of HEAs in high-temperature friction.

Hardness is among the most salient properties of numerous industrial materials. Consequently, a considerable body of research has recently emerged on the subject of HEAs, with a particular focus on the development of effective hardness prediction methods based on machine learning. Concurrently, these methods have demonstrated a commendable predictive capability, substantiated by experimental evidence, thus enhancing the efficacy of alloy design and reducing costs.

3.3. Strength prediction

HEAs are typically novel alloys composed of five or more elements. Due to the interaction between these elements, HEAs exhibit specific effects, such as lattice distortion and cocktail effects. These effects result in the development of exceptional properties, including high strength [29,175–181] high hardness and corrosion resistance. Strength is typically an essential performance metric for structural materials. In certain specialized applications, such as the manufacture of automobiles, aircraft, and ships, there is a demand for very high yield strength alloys as structural materials. Consequently, the accurate prediction of strength is paramount for alloys with stringent strength performance criteria.

Bhandari et al. [182] used the RFR algorithm of machine learning for the first time to predict the yield strength of HEAs at different temperatures. First, in order to describe other characteristics of the alloy such as VEC, etc., they are added to the data set by using formulas. Secondly, Pearson correlation coefficient and scatter plot were used to analyze the feature correlation. Finally, VEC and Tm were removed to obtain a dataset with 238 samples and 25 features. The data set is then divided into the training set and the test set in a ratio of 9:1. Moreover, the model is evaluated by the ten-fold cross-validation method, and MSE is used as the index of model evaluation. The results show that the prediction accuracy of RF on the training set reaches 95 %, which proves that the regression model has high precision. RF models predict the yield strength of MoNbTaTiW and HfMoNbTaTiZr at 800 °C and 1200 °C (Fig. 9a–b), respectively. However, the prediction accuracy of the former is higher than that of the latter, which may be related to the small number of data sets of the latter, so the use of larger data sets in the future may improve the accuracy. In addition, in order to verify the importance of features to the prediction results, density was deleted during the training process, and the result found that the prediction accuracy decreased, indicating that the selected features were important to the prediction results. In this study, the yield strength prediction of HEAs is realized with the help of the RFR model, which is in good agreement with the experimental results. However, the feature engineering does not include key microstructural parameters such as grain size, which may miss important influencing factors. In the future,

microstructural features and manufacturing parameters can be introduced as input variables to improve the prediction accuracy.

However, using only one model for prediction cannot guarantee the accuracy and reliability of prediction, so several common models should be compared to select the best model for prediction. Yan et al. [183] used four machine learning algorithms to optimize the model and used the optimized model to predict the yield strength. First, they collected a dataset of about 900 alloys from previous literature and selected 14 properties of the alloys as input characteristics. However, in order to avoid overfitting, 463 kinds of alloys were obtained after processing the original data, and 7 features were screened out by filtering method -Pearson correlation coefficient. The data set is then divided into a training set and a test set in an 8:2 ratio, and then four different algorithms, namely KNN, gradient boosting (GB), RF and DT, are used to make predictions, and the model is evaluated by the R^2 coefficient and RMSE. The prediction results show that GB algorithm performs best in all aspects of synthesis and is more stable. Therefore, the model's hyperparameters, namely estimators and max_depth, are optimized to perform best at 80 and 4 respectively, and the feature importance is ranked, ΔH_f is the most important feature related to intensity. Finally, in order to evaluate the generalization performance of the selected model applied to other alloys, the model was used to predict the data in the recently published literature, while predicted results were highly consistent with the experimental results. In this study, the dataset is dominated by single-phase alloys with limited coverage of elemental systems, which may not adequately reflect the strength mechanisms of multiphase alloys. It is recommended that physically embedded hybrid models be developed to capture the microevolutionary mechanisms and extended to strength prediction in multiphase systems.

In addition to the modeling algorithm, the selected features also have an important impact on the prediction effect and the prediction process. Zhang et al. [184] proposed a feature selection strategy to optimize the target features by borrowing the idea of AL in order to accelerate the screening of the component combination space. The feature pool of the initial HEAs has 255 features, and the subset of feature combinations is very large. To address this challenge, an AL strategy was adopted to select the most effective feature subset. Initially, the input features and training data were utilized to train a machine learning agent model. Then, the feature subset generated by the trained model was used as input, along with the autonomous learning engine, to search for optimal values in the data. The search results for high strength alloys were then validated through experimental testing. Training was initiated using the collected datasets, and four traditional methods were employed to select feature subsets for each set of data. Additionally, a gaussian process (GP) model was developed to predict strength in conjunction with the corresponding data. Subsequent results demonstrated that the effectiveness of the feature selection process exceeded that of the traditional methods (Fig. 9c–d). The selected features were then used to identify high strength alloys. After five experimental iterations, it was found that the strengths of the alloys obtained by this method exceeded those of the conventional method. In the fifth iteration, the yield strengths of the two alloys were 13.8 % and 21.3 % higher than the best values in the collected dataset, with the highest value reaching 3 GPa. The study innovatively uses AL efficiency as the feature selection criterion to discover ultra-high intensity HEAs efficiently, but the dependence of the AL strategy on small datasets may lead to insufficient model generalization in the complex component space, which can be enhanced by introducing a transfer learning strategy to enhance the robustness of the model in small sample scenarios by utilizing cross-system alloy data.

The enhanced strength of alloys can be attributed to various strengthening mechanisms, which can be modelled to facilitate the design of high-strength alloys. Li et al. [185] proposed a strategy for the design and mining of new alloys through the integration of machine learning and precipitation strengthening mechanisms (Fig. 10a). In previous studies, the inverse phase boundary energy and the volume

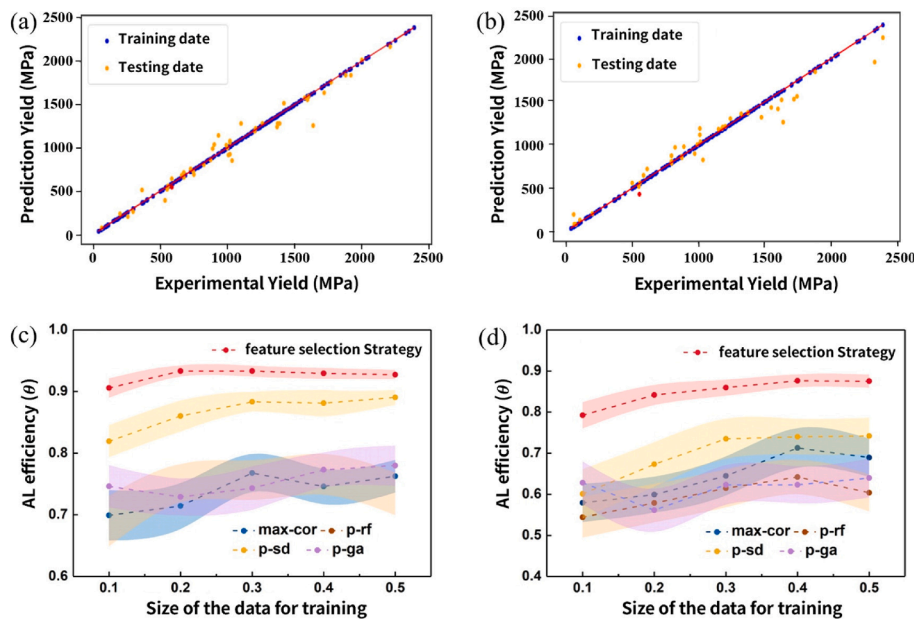


Fig. 9. (a,b) The yield strength prediction results of (a) MoNbTaTiW alloys and (b) HfMoNbTaTiZr alloys based on the RF model. Reproduced with permission from Ref. [182]. Copyright 2020, Materials Today Communications. (c,d) The plot of the AL efficiency results of the feature selection strategy proposed by the authors compared to the traditional strategy on the literature dataset (c) and the experimental dataset (d), respectively. Reproduced with permission from Ref. [184]. Copyright 2024, Journal of Materials Science & Technology.

fraction of the precipitated phase, which are associated with strength changes, were identified as two key parameters. Initially, they integrated the strengthening mechanism, machine learning and experimental synthesis, employing an automated machine learning framework to train the model. To validate the model's accuracy, they predicted and compared eight collected alloys, demonstrating that the predicted values were more aligned with the experimental results. Following the reduction of the search space, they identified 235,000 alloys from which they selected $\text{Al}_5\text{Ti}_8\text{Co}_{40}\text{Fe}_7\text{Ni}_{40}$ (Al_5Ti_8) for experimental investigation. A comparison was then made with $\text{Al}_7\text{Ti}_7\text{Co}_{40}\text{Fe}_6\text{Ni}_{40}$ (Al_7Ti_7), and it was found that the former has a higher anti-boundary energy but a smaller volume of precipitated phases. The tensile test results indicate that the yield strength of the Al_5Ti_8 alloy is 1.15 GPa, which is 10 % higher than that of the existing strengthened alloys. The findings demonstrate that the reversed-phase interfacial energy increases with the Ti/Al ratio, and that the strength of the alloys can be more effectively enhanced. In this study, an innovative fusion of precipitated reinforcement mechanisms and machine learning design HEAs successfully enhanced the yield strength, but the model relies on predefined reinforcement formulas, which may ignore the complex interaction effects of multiple elements. It is suggested that end-to-end machine learning models without a priori formulas can be constructed and then incorporate atomic scale parameters for high-throughput calculations.

In the context of alloys, the presence of a correlation between properties is a common occurrence. However, the prediction of a single property alone often falls short of the performance standards demanded by high-performance alloys. To address this shortcoming, He et al. [186] have developed a comprehensive framework for predicting the yield strength and fracture strain of RHEAs. This framework, built upon machine learning principles, serves as a significant advancement over the conventional trial-and-error approach. The framework is meticulously structured into four distinct segments: the construction of a database, the development of a model, the screening of components, and the experimental validation. The database is constructed by collecting data from published literature and experiments to form a dataset. The model is constructed as follows: firstly, a subset of nine optimal features is obtained by feature engineering. Subsequently, a grid search is used to optimize the hyperparameters of the model and a five-fold cross-validation is performed on the training set to ensure generalization ability. Among the

pre-selected models, RF exhibited the smallest RMSE and the largest R^2 , thus being selected as the yield strength prediction model. For strain prediction, the data were initially classified using a classification model, followed by a regression model (Fig. 10b–c). In the compositional screening stage, 792573 compositions were constructed and sent to the prediction model to select alloys with high yield strength and excellent fracture strain. Finally, $\text{Nb}_{0.18}\text{Ta}_{0.26}\text{Ti}_{0.22}\text{V}_{0.21}\text{W}_{0.13}$, $\text{Nb}_{0.26}\text{Ta}_{0.24}\text{Ti}_{0.21}\text{V}_{0.24}\text{W}_{0.05}$, $\text{Nb}_{0.22}\text{Ta}_{0.22}\text{Ti}_{0.24}\text{V}_{0.23}\text{W}_{0.09}$ and $\text{Nb}_{0.24}\text{Ta}_{0.22}\text{Ti}_{0.26}\text{V}_{0.04}\text{W}_{0.24}$ were selected, respectively. Experimental validation: the selected alloys were synthesized and tested, and the results showed that the model's accuracy of the yield strength and fracture strain prediction is still good. In this study, we successfully predicted the mechanical properties of RHEAs by integrating machine learning and feature engineering, however, we failed to achieve the synergistic optimization of high strength and high plasticity. The Pareto front of yield strength and fracture strain can be explored by introducing GA to design alloys with better overall performance.

Studying the mechanical properties of HEAs at high strain rates is also important for designing high-performance alloys. HEAs prepared by powder metallurgy are promising in the field of high-speed impact deformation. Li et al. [181] have investigated the mechanical behavior of CoCrFeMnNi HEAs prepared by powder metallurgy under room-temperature strains. Initially, powdered iso-atomic ratio CoCrFeMnNi HEAs were prepared, followed by a comprehensive evaluation of their properties. This involved a meticulous calculation of the associated stress, strain, and other pertinent parameters. Additionally, an in-depth analysis of the sawtooth behavior was conducted, employing state-of-the-art refined composite multiscale entropy (RCMSE) and mean field theory (MFT) techniques. The experimental results demonstrate that the alloys manifest sawtooth flow at elevated strain rates, with this phenomenon becoming increasingly pronounced as the strain rate rises. The yield strengths of the alloys at strain rates ranging from 1, 200s⁻¹ to 2,800s⁻¹ are between 500 and 700 MPa, which are higher than the values at lower strain rates. Furthermore, the alloys exhibit a high strain hardening rate. When the shear stress of the alloy reaches 635 MPa and the nominal strain is about 8.71, a shear zone of approximately 20 μm width is generated, which contains nanotwins and ultrafine equiaxed grains. The nanoindentation tests demonstrated that the strength of the shear zone was greater than that of the matrix, and

the strength exhibited a positive correlation with the strain rate. The RCMSE and MFT analyses demonstrated that the alloys prepared by powder metallurgy exhibited a milder sawtooth behavior and enhanced deformation stability in comparison to the cast and wrought alloys. Furthermore, the formation of nanotwin crystals, resulting from the synergistic effect of intense plastic deformation and rotational dynamic recrystallization (RDR), led to an enhancement in the strength of the alloys. This study elucidates the mechanical behavior and mechanical properties of HEAs that are prepared by powder metallurgy at elevated strain rates. And provides a theoretical foundation for the utilization of HEAs in the domain of high-speed impact deformation.

A plethora of studies have proposed various design methodologies for predicting the strength of HEAs. These methodologies have yielded favorable outcomes by employing a range of approaches, including model optimization, augmentation of the data set and feature subset, integration of the strengthening mechanism, and other techniques. In the future, advancements in these methodologies hold promise to enhance the accuracy of predictions and expedite the design process for high-strength alloys.

3.4. Thermodynamic properties prediction

In the context of HEAs, the consideration of thermodynamic properties is of paramount importance, in addition to mechanical and mechanical properties [30,175,187–190]. Critical properties such as lattice thermal conductivity, melting entropy, and martensite temperature are instrumental in elucidating the thermodynamic characteristics of HEAs. Conventional experimental or theoretical calculations are often time-consuming and costly. However, machine learning methods offer a potential solution, enabling the exploration of the vast composition

space of HEAs with greater efficiency and reduced cost. This approach provides substantial support for the design and development of new materials.

In high temperature environment, the influence of thermal deformation behavior on high strength materials is very significant. Because deformation will change the microstructure of the material, further affecting the performance. Therefore, in order to predict the thermal deformation behavior of HEAs, and reduce the forecasting work and cost. Sandeep et al. [191] used a machine learning algorithm to predict the flow stress of a $\text{CoCrCu}_{1.2}\text{FeNi}$ alloy with a double-sided heart-cube phase. First, they collected data sets from other people's literature and processed the data to remove outliers. At the same time, strain, temperature and strain rate are the input parameters, and flow stress is the output parameters. Then, the correlation between input and output parameters is analyzed by using correlation coefficient. Next, they used five machine learning algorithms: RF, KNN, XGB, DT and SVR to predict the flow stress of the HEAs (Fig. 11a). Grid search is used to optimize the hyper-parameters to minimize RMSE and maximize the prediction accuracy. After that, they divided the data set 8:1:1 into a training set, a test set, and a validation set. The MinMaxScaler was also used to normalize the data. Finally, the predicted results show that RF and KNN models perform best, R^2 is 0.987 and 0.986 respectively (Fig. 11b), and the predicted stresses are in good agreement with the actual observed stresses. In order to reduce the number of experiments and reduce the cost, the two optimal models are used to predict the thermal deformation behavior at 1,123 K temperature, and the results are good. Although this study verifies that the model performs well under conventional temperature/strain rates, the reliability of the prediction for extreme conditions needs further experimental verification. The physically-driven model can be combined with the data-driven model to optimize the prediction results using physical

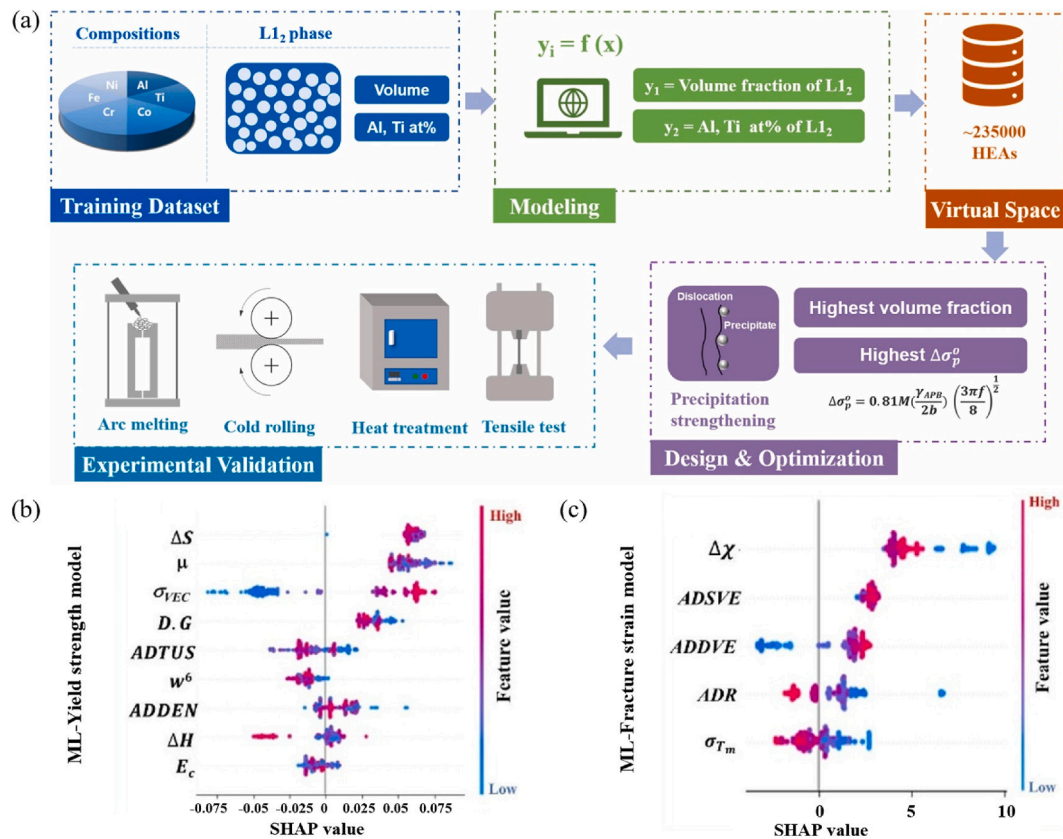


Fig. 10. (a) Flowchart for designing alloys combining reinforcement mechanisms, machine learning and experimental synthesis. Reproduced with permission from Ref. [185]. Copyright 2023, Materials Science & Engineering A. (b,c) The influence of different features on the SHAP values in yield strength model (b) and fracture strain model (c). Reproduced with permission from Ref. [186]. Copyright 2024, Materials & Design.

constraints, especially to improve the robustness under extreme conditions or data-scarce scenarios.

For high entropy shape memory alloys (HESMAS), martensitic phase transition temperature (M_p) and thermal hysteresis (T_{hy}) determine their application potential. In order to develop the new HESMAS, M_p and T_{hy} need to be designed. Tian et al. [150] used machine learning methods to construct a design process for efficient design of alloys. First, they collect data from the literature and form a dataset. Then, two-step feature screening was used for the generated features. Pearson correlation coefficient was used first, and Boruta hap model was used to select the best combination. Since M_p and T_{hy} are continuously variable, they used five regression models: linear regression (LR), decision tree regressor (DTR), RFR, extreme gradient boosting regressor (XGBR), and light gradient boosting machine (Lgbm). They then divided the original data set into a training set and a test set on an 8:2 basis. The above five models are used to predict on two datasets, and the results show that XGBR has the best performance. In addition, to improve the predictive performance of the model, they used BO to optimize the model hyperparameters. The optimized model performs well on both M_p and T_{hy} datasets. To verify the generalization ability of the model, they evaluated it using leave-one-out cross-validation (LOO-CV), with final R^2 values of 0.749 and 0.814 (Fig. 11c–d), respectively. To analyze the impact of features on targets, they interpreted the model using SHAP, where Hf had the largest impact on M_p and Cen had the largest impact on the model. In T_{hy} , Ti has the greatest influence on the model, while Cen has the greatest influence on T_{hy} . Finally, by applying the constructed model, the simultaneous prediction of M_p and T_{hy} can be achieved, so as to screen the desired target HESMAS. The study achieves accurate prediction of complex phase transition behavior through a dual-objective machine learning framework and SHAP interpretation. However, the combination of dataset elements is single, and it is recommended to broaden the range of dataset gold system.

The martensitic transition temperature (Ms) is a critical property of shape memory alloys; however, experimental measurements of Ms are currently time-consuming and laborious. To address this need, Liu et al.

[192] constructed a model to predict the relationship between alloy composition and Ms using a machine learning approach. The model was built using data collected in the literature, with Ms as the target variable and alloy elemental composition as the raw feature. The raw features were then transformed into physical descriptors, and the linear correlation of the features was evaluated using PCC. Subsequently, they used recursive feature elimination (RFE) to remove unimportant features and selected six key features using an exhaustive search method, with $\Delta\chi$ demonstrating the most significant impact on SHAP values (Fig. 12a). In selecting the model, they compared four different algorithms: gradient boosting tree (GBT), bagged regression (BAR), SVR, and KNN. Performance was evaluated using R^2 and MAE. The results demonstrate that the GBT model exhibits the highest level of prediction accuracy and effective generalization ability. For conventional Ni-Ti memory alloys, the relatively low Ms limits their applications. However, the process of doping with additional elements poses challenges, such as high cost and a broad compositional space. To address these limitations, a TiZrHfNi-CuCo high-entropy shape memory alloy was designed by combining the concept of HEAs. The constructed prediction model was utilized to screen the alloy composition, which led to the improvement of the Ms value of the alloy. The results show that the Ms value of the Ti₁₉Zr₁₉Hf₁₉Ni₃₇Cu₆ alloy is higher than 400 °C (Fig. 12b), a finding that suggests improved performance of the alloy for high-temperature applications. Although this study HEAs design is effective in enhancing Ms, the use of precious metals in the composition increases the cost and environmental compatibility was not evaluated. In the future, low-cost elemental alternatives should be explored to balance alloy performance with economics and environmental friendliness.

For HEAs, thermodynamic stability is essential for their application at high temperatures. However, the traditional descriptors have limitations, so Zhang et al. [193] introduced two characteristic temperatures as new descriptors. Firstly, 1980 structures obtained by enumeration structure search and filtering are used for training, the energetic models of cluster expansion (CE) Hamiltonian of the body-centered cubic Ti-Zr-Hf-Nb-Ta alloy was optimized (Fig. 12c), and least absolute

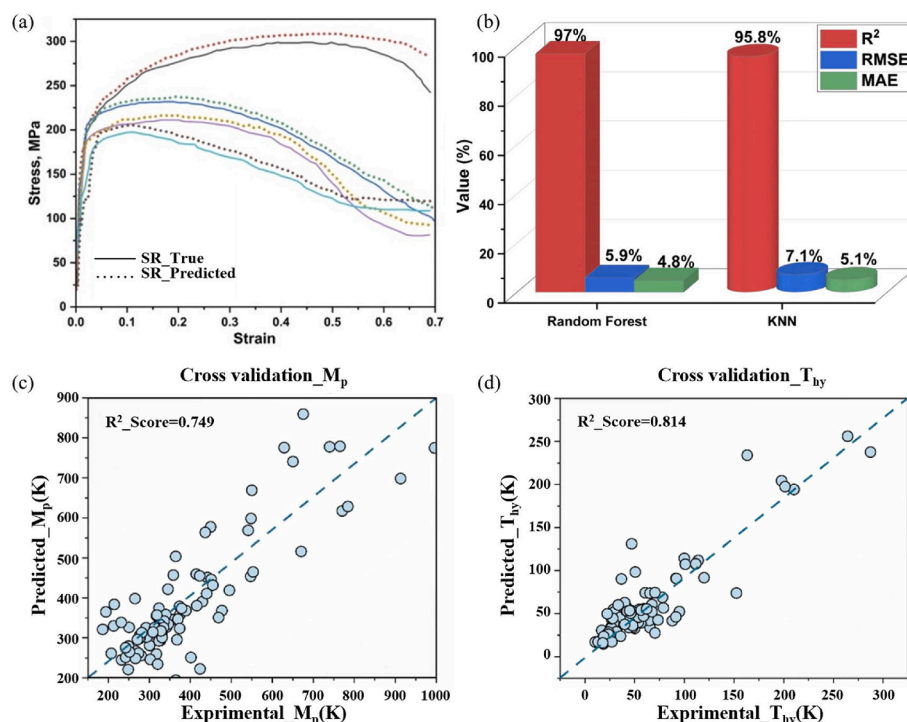


Fig. 11. (a) Plot of the results of strain rate prediction using the RF model at the new temperature of 1123 K against the experimental results and (b) Plot of performance metrics of RF and KNN models in predicting flow stresses at new temperatures. Reproduced with permission from Ref. [191]. Copyright 2024, Materials Today Communications. (c,d) The plots of the final R^2 score results obtained by LOO-CV for the (c) M_p and (d) T_{hy} models, respectively. Reproduced with permission from Ref. [150]. Copyright 2024, Materials Today Communications.

shrinkage and selection operator (LASSO) regression was used to avoid overfitting. Then, they calculate the configuration density (cDOS) of the alloy by using the Wang-Landau random walk method, and then obtain the distribution function. Then configurational entropy, free energy and internal energy are determined. It can be seen from the prediction results that CE model prediction is highly correlated with density functional theory (DFT) calculation results, and $R^2 = 0.94$ (Fig. 12d), indicating that the optimized model is highly accurate. Then the alloys of Ti-Zr-Hf-Nb-Ta system were studied, and it was found that the cDOS of binary alloys showed the separation of stable and unstable components, while the cDOS of ternary alloys and above overlaps and the distribution becomes narrow. At low temperature, high element alloys maintain metastable state due to the decrease of configurational entropy, so the stability cannot be judged only by ideal mixing entropy. They then described the entropy enhancement ratio by introducing $Q_0(N_{spe})$ and $C(x, T)$. And the new criterion $C(x; 300K)/Q_0(N_{spe}) - 1 > 0$ has better classification effect than other descriptors. Then, two characteristic temperatures, T_{bo} and T_{in} , are defined to distinguish the alloys with different stability, and the effect is remarkable. This study proposes to construct a new descriptor and a new alloy classification method, which breaks through the entropy limitations of the traditional configurations and can be extended to more alloying systems to validate the universality of the descriptors.

The thermodynamic properties of HEAs materials have a significant impact on their design and application. A substantial body of research has been dedicated to the development of alloys with specific properties through diverse methodologies. These methodologies include the construction of predictive models and the utilization of machine learning potentials or machine learning interpotentials to predict thermodynamic properties and behaviors. These studies underscore the significant potential of machine learning algorithms in accelerating the design and development of HEAs.

3.5. Catalytic properties prediction

Presently, global energy consumption is on the rise, and the environmental challenges associated with it are becoming increasingly pronounced. The pressing imperative, therefore, is to devise effective methods for the conversion and storage of clean and sustainable energy. The utilization of catalysis holds promise in addressing this issue by facilitating the splitting of stored water into hydrogen and oxygen, thereby enabling the efficient use of renewable energy sources. Concurrently, catalysis can address the issue of carbon dioxide emissions, thereby facilitating the realization of carbon neutrality. Metal catalysts, particularly HEAs, have emerged as a particularly promising class of catalytic reagents, exhibiting superior catalytic properties when compared to traditional alloys [31,45,194–197]. Furthermore, machine learning has emerged as a pivotal instrument in expediting the identification of catalysts for HEAs.

In order to design efficient catalysts and to bypass the scalar relationship between the adsorption energies of carbon dioxide reduction reaction (CO₂RR) intermediates. Chen et al. [198] placed Fe_{0.2}Co_{0.2}Ni_{0.2}Cu_{0.2}Mo_{0.2}(111) surface as the active surface for the CO₂ reduction reaction, 200 structures were generated. And 20 structures were selected to calculate the adsorption energy of their intermediates, and finally 1280 adsorption sites were used as a database. Then they created three NN models to predict the adsorption energies of the intermediates using five relevant features as inputs. The prediction results indicate that the adsorption energy follows a Gaussian distribution, and the adsorption energy values predicted by the model demonstrate a high degree of agreement with those calculated by DFT. By analyzing the significance of the features, they found that the scalar relationship could be broken by adjusting the chemical environment of the active site. Subsequently, by analyzing the model, they proposed that the location with the greatest influence on the adsorption energy is the top site. And different surface correlations are different, which provides the possibility to break

the scaling relationship. Therefore, they chose Co, Fe and Ni as the active atoms in the AS1-3 centers, and Mo and Cu as the surface-adjacent atoms, respectively, which made the proportionality between the CO and COOH (CHO*) (Fig. 13a) adsorption energies disappear. Finally, the CO₂RR process was calculated for the AS1-3 centers, which resulted in better catalytic performance compared to the reported values. Moreover, the proportionality between COH* and CO* disappeared, which further verified its high catalytic performance. But the study focuses on idealized crystal surfaces and ambient temperature and pressure conditions, and does not consider the effects of the actual electrocatalytic environment on intermediate adsorption and reaction pathways. In the future, electrolyte effects and high-pressure conditions can be integrated in the model to enhance the generalization of the catalytic performance.

In order to explore efficient catalysts quickly, predict catalytic properties stably and study complex catalytic mechanisms. Shan et al. [199] mined high-efficiency and HEAs electrocatalysts by combining data-driven strategies and high-throughput calculations (Fig. 13b). First, they combined large language model (LLM) with the Language Prompt project to classify the databases extracted from Clarivate Web of Science. Moreover, two databases of catalytic reaction and hydrogen evolution reaction (HER) were established, in which nine key elements were counted and Sn was also included. They used PtPd-based five-element HEAs as HER model and generated 56 combinations. The mixed metal salt precursor solution is then printed on the glass carbon (GC) surface using a high-throughput printing workstation. Then the HEAs array can be prepared by rapid pulse heating, and its microdot size can be controlled. Moreover, the elements of the synthesized sample are evenly distributed and the surface structure is good. Scanning electrochemical cell microscopy was used to evaluate HER activity of the catalyst, and the relative current density ($j_{relative}$) was introduced to characterize the intrinsic activity of the catalyst. They screened a comprehensive set of 56 active datasets and Fe-Co-Ni-Pt-Pd systems. Subsequently, a model was used to predict the active datasets and recommend two main sets of ingredients. In order to verify the accuracy of the model, the recommended components were prepared macroscopically. X-ray diffraction (XRD) characterization shows that the element distribution of the sample is uniform and the structure is good. And sample 1 (Fe_{0.15}Co_{0.40}Ni_{0.05}Pt_{0.32}Pd_{0.08}) in the overpotential, and Tafel slope stability index are outstanding, ensemble machine learning (EML) model has accuracy and practicability. The study focuses on HER and the elemental combination is limited to PtPd-based systems, which is not sufficiently applicable to other catalytic reactions. It is suggested that it can be extended to multiple reaction types such as CO₂RR and a multi-objective optimization model across reactions can be introduced.

Most of the existing catalytic studies are still limited to the low-dimensional compositional space, and HEAs are expected to be novel and efficient catalysts due to their unique active site distribution. However, existing single-target BO methods have limitations, and Xu et al. [200] instead constructed a multi-target BO framework for exploring catalysts for oxidation-reduction reactions (ORR) (Fig. 14a). The design space is mapped to the performance space through agents and empirical models based on DFT data. However, the calculation of adsorption enthalpy is very difficult due to too many active sites of HEAs. Therefore, they used the GNN model to predict the adsorption enthalpy and integrated it with the heuristic current density modelling technique to predict the catalytic activity (Fig. 14b). The test results showed small MAE for adsorption enthalpy prediction and good prediction performance on out-of-domain datasets, indicating the extrapolation capability and usefulness of the model. In addition to catalytic activity, there is another objective: cost-effectiveness. By performing a bi-objective BO optimization in the HEAs space, it was found that the Pareto front was wider for AgIrPdPtRu and AgCuIrPdRe. And the dual-objective approach can provide low-cost and high-activity candidate materials. In addition to the above two objectives, there is another objective: mixing entropy. Similarly, after dual-objective optimization, it is known that it usually conflicts with activity, but AgIrPdPtRu and

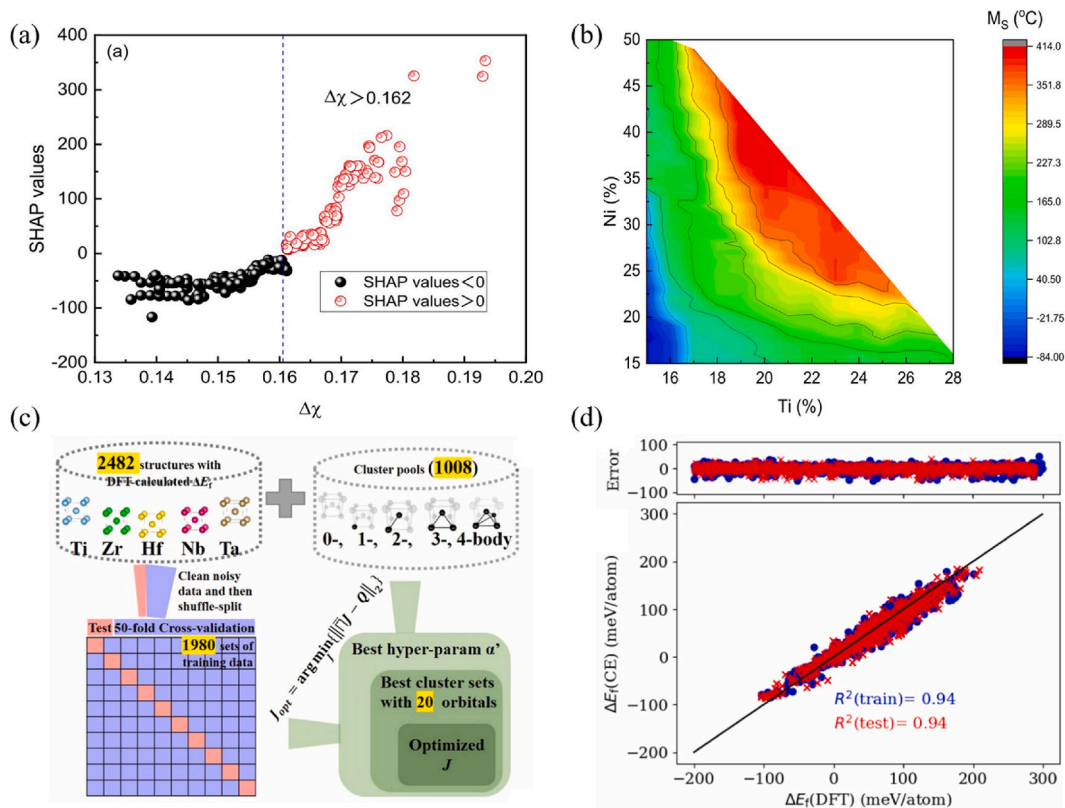


Fig. 12. (a) Plot of the relationship between $\Delta\chi$ feature values and SHAP values; (b) Contour plots of the effect of elemental composition on M_s values in alloys. Reproduced with permission from Ref. [192]. Copyright 2024, Materials Today Communications. (c) Flowchart for constructing a CE model by using a DFT dataset; (d) Differences in formation energy between DFT and CE calculations and comparison of R^2 on test and training sets. Reproduced with permission from Ref. [193]. Copyright 2024, Acta Materialia.

AgCuIrPdRe are still more promising. Finally, they performed a three-objective optimization of the 10-element HEAs space and found that many Pareto-optimal solutions are inside the frontier. Although this study enhanced the comprehensiveness of HEAs electrocatalyst screening through multi-objective optimization, the objectives only included activity, cost and mixing entropy, ignoring the key practical factors such as stability, which can be followed up by incorporating the stability, synthesis feasibility and other multivariate indexes to improve the objective function.

In order to screen highly active HEAs catalysts in a broad compositional space and to optimize the alloy composition, Clausen et al. [201] designed a computational framework to accelerate the development of HEAs catalysts. First, they used DFT to sample the binding sites on the surface of HEAs and used the generalized gradient approximation-density functional theory (GGA-DFT) method to calculate the adsorption energy. To train the regression model, the components of each plate were sampled from a Dirichlet distribution ($\alpha = 1$), and a graph convolutional neural network (GCN) was used to predict the adsorption energy. Subsequent to determining the constituent metal molar ratios, the face-centered cubic (111) surface of a HEAs of $N \times N \times 3$ ($N = 96$) was simulated, and the adsorption energy was labeled using a regression algorithm to obtain the total adsorption energy distribution. It is considered that the adsorption free energy and potential relationship, as well as adsorbate interactions, are the factors that limit adsorption behavior. To this end, they methodically introduced adsorbates in a stepwise manner to ascertain the net adsorption energy distribution. Subsequently, they derived a kinetic limiting current expression, underpinned by the redox step and the Gibbs reaction energy, which facilitated the calculation of the catalytic activity of the alloy. Ultimately, the predicted catalytic activity of the Ag-Pd edge was identified as the most optimal within the five-element composition space through the optimization of the composition space by means of

coarse network or BO. This study presents a flexible computational framework and argues for a comprehensive understanding of the key role of factors such as net adsorption energy distribution. However, it only focuses on the single-objective optimization of catalytic activity, and the usefulness of the prediction can be improved by introducing a multi-objective optimization framework and extending it to higher dimensions of the alloy space.

The exploration of high-entropy metal oxides (HEOs) has proven to be a challenging endeavor for researchers, particularly when employing conventional trial-and-error methodologies. This is attributed to the intricate compositional combinations that characterize these materials. To address this challenge, Duan et al. [202] adopted a machine learning approach to predict the catalytic oxidation properties and crystallographic behavior of a vast array of HEOs spaces (Fig. 15a). The experimental setup involved the preparation of 381 samples. The crystal structures were observed by XRD, and the microstructures and elemental distributions of the catalysts were analyzed by transmission electron microscopy (TEM) and energy dispersive spectroscopy (EDS). The activity and stability of the catalysts were then tested, with the catalytic oxidation activity measured by T_{90} (i.e. the temperature at which 90 % of the reactants are converted). Subsequently, an elemental composition-crystallization classification model (Fig. 15b) and an elemental composition-catalytic activity regression model were constructed by evaluating the models using a five-fold cross-validation method (Fig. 15c) and adjusting the hyperparameters using BO. The evaluation results showed that the naive bayes algorithm (NB) had the best overall performance in the classification model and the GBR algorithm had the best performance in the regression model. The improved GBR model was then used to screen samples with low T_{90} . For CH_4 oxidation, they combined crystal label prediction and T_{90} screening to select $\text{Ni}_{0.04}\text{Co}_{0.48}\text{Zn}_{0.36}\text{V}_{0.12}\text{Cr}_2\text{O}_x$ samples with the lowest T_{90} (365

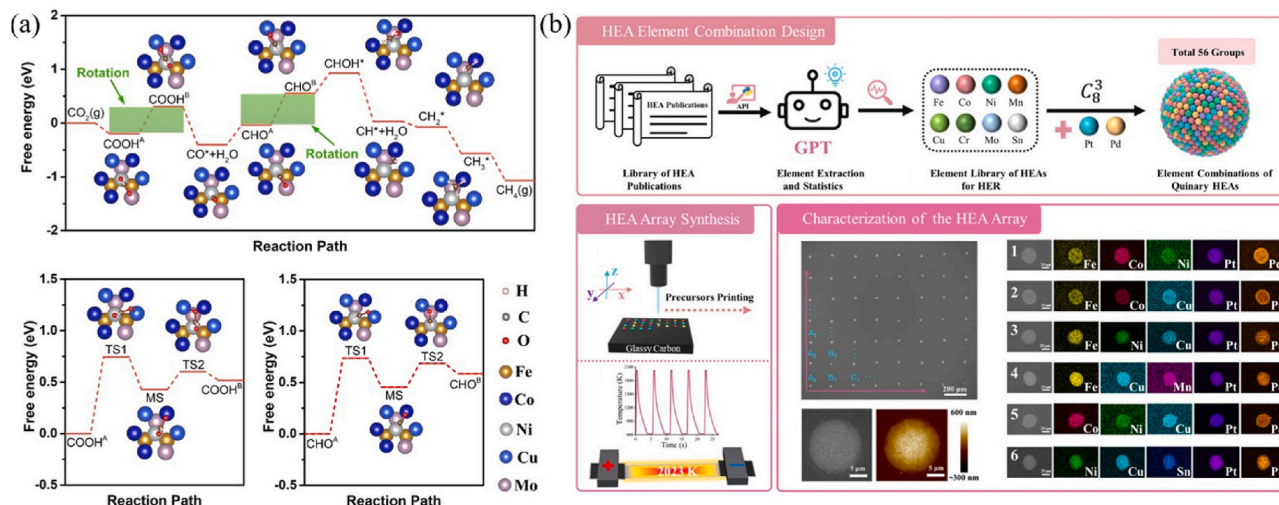


Fig. 13. (a) The CO₂RR catalyzed reaction process and the rotation of COOH* and CHO* on AS1. Reproduced with permission from Ref. [198]. Copyright 2022, ACS Catalysis. (b) The design, synthesis and characterisation process of microscale HEAs catalyst arrays. Reproduced with permission from Ref. [199]. Copyright 2024, Nano letters.

°C). The HEAs catalysts, in addition to their high catalytic activity, also have excellent stability. For instance, Ni_{0.04}Co_{0.48}Zn_{0.36}V_{0.12}Cr₂O_x demonstrates high oxidation activity and stability in CH₄, along with good tolerance to SO₂ and H₂O, making it a promising candidate for industrial applications. This study efficiently screens HEAs catalysts by machine learning. However, the training set of the study is not diverse enough and lacks an in-depth explanation of the microscopic catalytic mechanism. It is recommended to expand the model to a wider range of

element combinations and validate the synergistic effects and active sites between the elements by computational methods such as DFT to enhance the interpretability of the model.

Recent advancements in the field have yielded encouraging outcomes, particularly in the realm of investigating the catalytic capabilities of HEAs. Concurrently, machine learning has emerged as a novel avenue for the investigation of efficient HEAs catalysts. The employment of machine learning has expedited the identification of potential HEAs

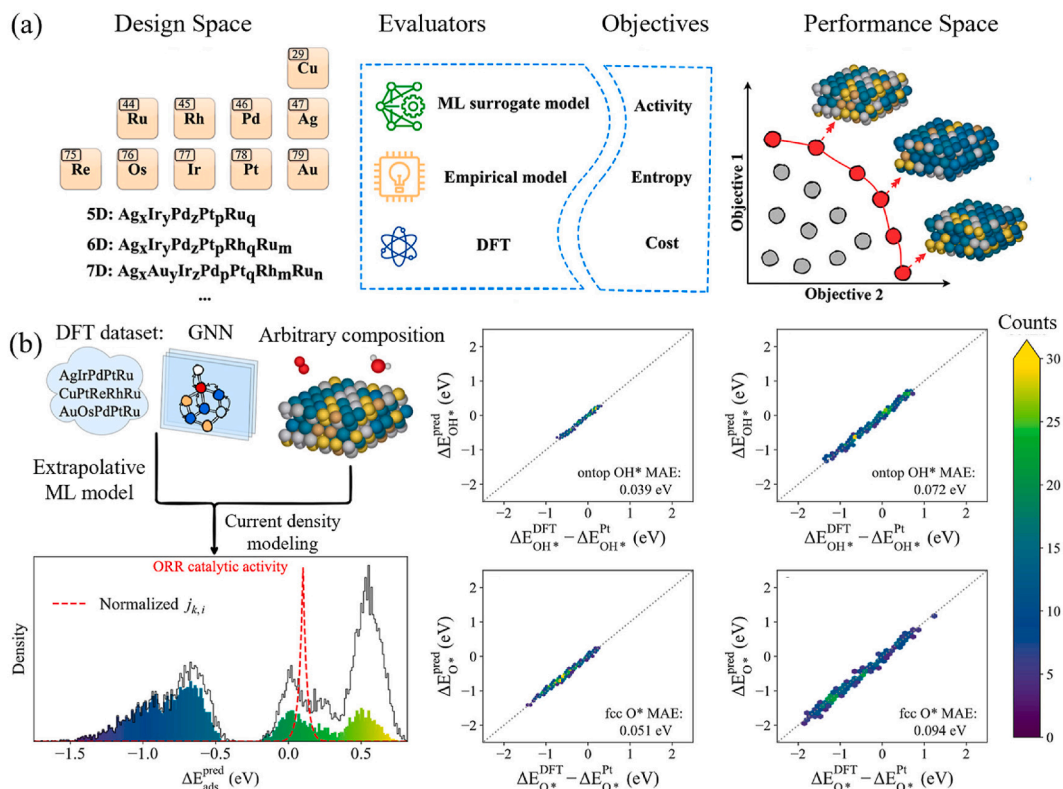


Fig. 14. (a) Design framework diagram for HEAs electrocatalysts for ORR; (b) Schematic diagram of catalytic activity prediction using machine learning, and parity plots of machine learning prediction and DFT calculation of adsorption enthalpies on two out-of-domain datasets. Reproduced with permission from Ref. [200]. Copyright 2024, Journal of the American Chemical Society.

that exhibit exceptional catalytic capabilities. However, challenges persist, including the interpretability of models and the integration of theory and practice. In the future, it is imperative to continue developing relevant algorithms and visualization tools to facilitate a transition from theoretical exploration to practical application.

4. Conclusions and outlook

4.1. Conclusion

In this review, we provide machine learning-based insights into the microstructure and performance of HEAs. We begin with an introduction to machine learning, focusing on the process of machine learning: data collection and processing, feature selection, model selection, model construction and prediction, as well as the methods applied in each process. For example, eight algorithms such as SVM, RF, KNN, ANN and GAN are introduced for algorithm selection. After that, the applications of machine learning in the prediction of microstructure and properties of HEAs are listed. This mainly includes phase composition, hardness, strength, thermodynamic properties and catalytic properties. Through these performance prediction results, we can see that machine learning has played a great role in promoting HEAs research, which is mainly reflected in the fact that machine learning is more efficient in handling and processing a large amount of HEAs data, which can reduce a lot of time and cost of manual trial and error. And it can explore potential new materials in the huge compositional space, which provides new ideas and new opportunities for future HEAs research.

Moreover, we also summarize the key features of each target prop-

erty and the algorithms with better accuracy and overfitting resistance. In phase structure prediction, the key features are valence electron concentration VEC , δ , ΔH , ΔS , T_m , $\Delta\chi$, RF is better against overfitting, and SVM is more accurate. In hardness prediction, the key features are VEC , δ , ΔS , E_f^{3rd} , ΔH , T_m , RF algorithm has higher accuracy, XGBoosting has better resistance to overfitting, integrated model consisting of ANN, KNN and LR have better accuracy and anti-fitting ability. In strength prediction, the key features are ΔS , ΔH , T_m , δ , lattice distortion energy, GB has better accuracy and RF has better overfitting resistance, and the combined two metrics of GB algorithm performs more robustly. In the prediction of thermodynamic properties, the key features are ΔS , ΔH , T_m , M_s and $\Delta\chi$, XGBR has higher accuracy and RF has better overfitting resistance, and GBT has both better prediction accuracy and overfitting resistance. ability is better. In the prediction of catalytic performance, the key features are adsorption energy, catalytic activity, adsorption enthalpy, and distribution of surface-active sites, GBR algorithm has higher accuracy, GNN has better resistance to overfitting.

4.2. Outlook

Despite the great potential of machine learning in the field of HEAs research. And with its help, the research on HEAs has made great progress. However, the research of HEAs still faces many challenges, and its future direction can be developed in the following aspects:

Scarcity of multicomponent metadata and high-dimensional feature space: HEAs composition space contains more than five elements, and its composition space grows exponentially, leading to extreme scarcity of experimental and computational data, resulting in a

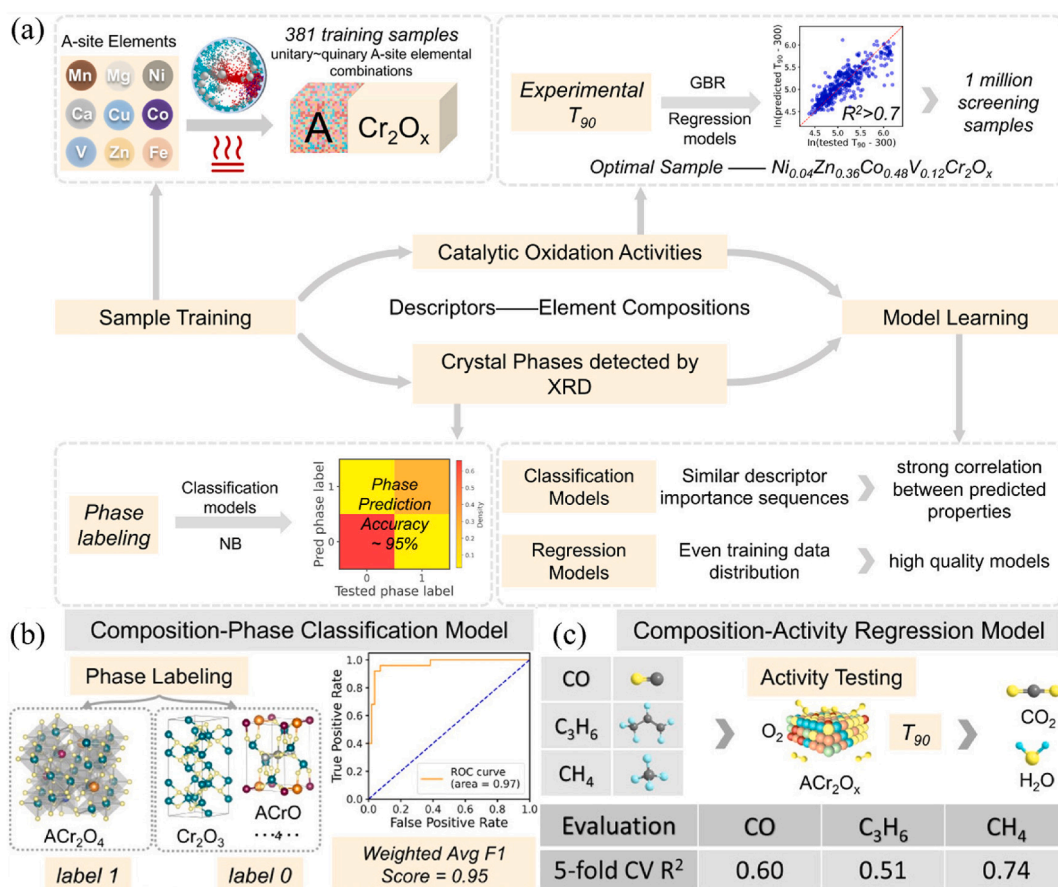


Fig. 15. (a) The scheme for predicting the crystalline phase structure and catalytic properties of ACr_2O_x systems using machine learning methods; (b) The labelling strategy for combining classification models and the results of predicting validation samples; (c) The catalytic schemes and 5-fold cross validation (CV) R^2 scores for CH_4 , CO , C_3H_6 based on catalytic activity regression models. Reproduced with permission from Ref. [202]. Copyright 2024, Journal of the American Chemical Society.

dimensional disaster. Traditional alloy synthesis is time-consuming and labor-intensive, and making it difficult to support the construction of data sets for thousands of samples. In addition, the feature space of HEAs contains primitive, plus the dimensionality of derived features often exceeds 50, and the performance of traditional machine learning models degrades drastically with increasing dimensionality. We can combine high-throughput experiments with the AL algorithm to synthesize hundreds of samples in parallel using a robotic experimental platform. Redundant data collection can be reduced by combining the AL algorithm to dynamically assess model uncertainty and prioritize the most informative samples for experimentation. In addition, constructing open shared databases and developing data standards for HEAs can enhance the ability to migrate models across systems.

Complex physical mechanisms and lack of a priori knowledge:

The unique physical mechanisms of HEAs, such as high entropy effect, lattice distortion, and cocktail effect, have not been fully quantified. Although the mixing entropy is widely used, it only describes the statistical disorder and cannot reflect the effect of atomic scale localized structure on the performance; the difference in atomic size of multi-component atoms leads to lattice distortion, triggering the localization of electrons, which affects diffusion barriers and electrical conductivity, but it is difficult to correlate the atomic scale distortion with the macroscopic performance in the machine learning model. we can develop descriptors for HEAs, and at the same time, combine traditional descriptors to construct a hierarchical feature space to improve the predictive ability of the model. Moreover, we can integrate multi-scale models by dividing the work at different layers, thus forming a cross-scale prediction framework of atomic scale-microstructure-macroscopic performance.

Model interpretability and physical mechanism consistency:

HEAs Complex component-structure-property mappings disconnect the results of machine learning modeling from physical mechanisms. For example, SHAP value tools can identify key features, but it is difficult to answer why a feature is important or how to modulate composition to improve performance. And it is difficult to establish a quantitative relationship between both model output and microstructure to guide alloy design. Dynamic processes such as catalysis and deformation involve time-scale effects, and existing static models cannot capture the dynamic pathways. To solve these problems, we can jointly analyze SHAP and DFT, first using SHAP values to quantify the characteristic contributions and identify key parameters, and then using DFT calculations to verify the physical mechanism basis. In addition, if the prediction results do not conform to the physical laws, a penalty term can be added to the model training to increase the loss value and force the model to conform to the physical laws.

Interdisciplinary collaboration and technology integration barriers:

The lack of interoperability of expertise between materials scientists and machine learning experts leads to model design that does not meet the needs of actual HEAs. Moreover, the format of experimental data and simulation data is not unified, and the screening of experimental data requires manual transformation, which is time-consuming and easy to introduce errors. Existing studies are mostly single prediction and lack the feedback mechanism of experimental validation-model update. After experimental validation, no new data are injected back into the model, resulting in no improvement in the accuracy of subsequent predictions, making it difficult to construct a closed-loop optimization process. In the future, we can integrate an automated platform to realize the closed loop of alloy design-synthesis-testing-model update. In addition, machine learning toolkits with built-in HEAs data preprocessing modules, specialized algorithms, and database interfaces can be developed to simplify operations.

Looking ahead, although research on HEAs has made progress in most fields, applications in some extreme environments such as aerospace, nuclear reactors, and biomedicine and other specific functional areas are still lacking, and future focus can be on the needs of these areas to promote the leap from theory to practical applications of HEAs. In

addition, the research should also respond to the dual-carbon goal, reduce the use of precious and toxic metals, improve the recycling rate, and form a multi-objective optimization paradigm of performance-cost-environment.

CRediT authorship contribution statement

Xiaoyi Zhang: Validation, Writing – original draft, Data curation, Investigation, Formal analysis. **Wenhan Zhou:** Funding acquisition, Writing – review & editing, Supervision. **Xiang Li:** Writing – review & editing, Formal analysis. **Tong Xu:** Formal analysis, Writing – review & editing. **Yongzhen Yu:** Writing – review & editing. **Lei Zheng:** Writing – review & editing. **Guanhua Jin:** Supervision, Writing – review & editing. **Shengli Zhang:** Writing – review & editing, Funding acquisition, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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